Flying Lab for Airborne Isotope Research

ASEN 6028 Spring 2026

Operational Readiness Review

Technical Report



Ann and H.J. Smead Aerospace Engineering Sciences Department,

The University of Colorado Boulder, Boulder, CO, 80309

April 24, 2026

Table of Contents

List of Acronyms/Abbreviations 8
List of Symbols 10
Chapter 1: Mission Overview and ORR Objectives 12
1.1 Executive Summary 12
1.2 Mission Objectives and Success Criteria 15
1.3 Mission Schedules 16
1.4 Work Breakdown Structure 18
1.6 Flight Readiness Criteria 19
Chapter 2: Concept of Operations 20
2.1 Operational Mission Profile 21
2.2 Pre-Flight Operations 22
2.3 In-Flight Operations 22
2.4 Post-Flight Operations and Data Handling 22
2.5 System Requirements 22
Chapter 3: Final System Configuration 24
3.1 Payload Overview/Updates 24
3.1.1 Mechanical Updates 25
3.1.2 Electrical Updates 27
3.2 Mechanical Configuration (As-Built) 28
3.3 Electrical Wiring and Connections 29
3.3.1 Power Distribution 29
3.3.2 Control System (Arduino Nano) 29
3.3.3 Sensor Interface (HMP60) 30
3.3.4 Actuator Interfaces 30
3.3.5 Logic Level Shifter 30
3.3.6 Communication Interface 31
3.3.7 Grounding Strategy 31
3.3.8 Wire Types and Connections 31
3.4 Electrical Architecture (v3 = Final) 31

3.4.1 System Overview **32**
3.4.2 Power Architecture **33**
3.4.3 Control Architecture **33**
3.4.4 Design Characteristics **33**
3.5 Payload Software and GUI Configuration **33**
3.5.1 Payload Software **33**
3.5.2 GUI Configuration **35**
3.5.3 Upgrade 1: Raw Packet Display (Hexadecimal Diagnostic View) **37**
3.5.4 Upgrade 2: Fan Speed Control from Ground Station **37**
3.5.5 Upgrade 3: Offline Post-flight Analysis Mode **37**
3.6 Mass Properties and Center of Gravity Verification **38**
3.7 Airflow Configuration **40**
3.7.1 Airflow Overview **40**
3.7.2 Inlet **40**
3.7.3 Honeycomb Flow Straightener **41**
3.7.4 Turning Vanes **42**
3.7.5 Fan & Exhaust **43**
3.7.6 CFD Analysis **44**
3.7.7 Testing **47**
3.7.8 Fan Control Implementation **48**
3.7.9 Future Outlook **49**
Chapter 4: Integration and Assembly **50**
4.1 Fabrication Summary **50**
4.2 Mechanical Assembly Process **51**
4.2.1 Laser and Mirror Mount Assembly **51**
4.2.2 Enclosure Installation **52**
4.2.3 Inlet Assembly **52**
4.2.4 Exhaust Assembly **54**
4.2.5 Bulkhead Installation **55**
4.2.6 Sealing and Vibration Control **56**
4.2.7 Final Inspection **57**

4.3 Electrical Integration 57
4.3.1 Sensor Installation 57
4.3.2 Laser Electronics Installation 57
4.3.3 PCB Assembly & Installation 57
4.3.4 Final Electrical Connections 58
4.4 Payload Integration into Black Swift S3 60
4.5 Integration Challenges and Resolutions 60
Chapter 5: Verification and Testing 61
5.1 Electrical Testing Results 61
5.1.1 Sensor Characterization (HMP60) 62
5.1.2 Fan Control Characterization 62
5.1.3 Signal Interface Validation (Laser Integration) 62
5.1.4 Power System Validation (Integrated Operation) 63
5.1.5 Full System Integration with Drone 63
5.2 Mechanical Testing Results 63
Chapter 6: Budget Analysis 67
6.1 Cost Budget Analysis 67
6.2 Mass Budget Analysis 68
6.3 Volume Budget Analysis 69
6.4 Power Budget Analysis 70
Chapter 7: Communications and Ground Operations 71
7.1 Telemetry Architecture Verification 71
7.2 GUI Operational Validation and Data Storage Verification 72
Chapter 8: Risk Closure and Safety 72
8.1 Updated Risk Assessment 72
8.2 Risk Retirements Since CDR 74
Chapter 9: Conclusions and Path Forward 78
9.1 ORR Outcomes 78
9.2 Remaining Work 78
9.2.1 Payload Software 78
Chapter 10: References 80

List of Figures

FIG. 1 PDR SCHEDULE. 16
FIG. 2 CDR SCHEDULE. 17
FIG. 3 ORR SCHEDULE. 18
FIG. 4 FLAIR WORK BREAKDOWN STRUCTURE. 19
FIG. 5 CONCEPT OF OPERATIONS. 21
FIG. 6 EXPANDED CONCEPT OF OPERATIONS. 21
FIG. 7 BULKHEAD CHANGES, BEFORE (A) AND AFTER (B). 25
FIG. 8 DIAGRAMS OF ELECTRONICS MOUNTING, LASER MOUNTING(A), PCB MOUNTING (B). .. 26
FIG. 9 MODIFIED ENCLOSURE CAD DIAGRAM. 26
FIG. 10 ENCLOSURE COVERED WITH TAPE. 27
FIG. 11 PAYLOAD MOUNTING WITH TAPE. 27
FIG. 12 FULL ASSEMBLY CAD MODEL. 28
FIG. 13 ASSEMBLED PAYLOAD. 29
FIG. 14 FINAL ELECTRICAL SCHEMATIC. 32
FIG. 15 FINAL PCB LAYOUT. 32
FIG. 16 GUI CONFIGURATION. 36
FIG. 17 PACKET DISPLAYED IN HEX STREAM FORMAT. 37
FIG. 18 MASS BUDGET. 38
FIG. 19 CDR CENTER OF GRAVITY. 39
FIG. 20 DAY BEFORE FLIGHT CENTER OF GRAVITY 39
FIG. 21 AIRFLOW SUBSYSTEM ISOMETRIC VIEW. 40
FIG. 22 INET TOP AND ISOMETRIC VIEW. 41
FIG. 23 HONEYCOMB FRONT AND ISOMETRIC VIEW. 42
FIG. 24 HONEYCOMB FLOW CONDITIONING. 42
FIG. 25 TURNING VANES IN AIRFLOW SUBSYSTEM. 43
FIG. 26 SAN ACE 40 9HVA IN AIRFLOW SUBSYSTEM. 44
FIG. 27 FAN CURVE INTERPOLATION. 44
FIG. 28 FRONT AND ISOMETRIC VIEW OF CFD BOUNDARY CONDITION LOCATIONS (RED). 45
FIG. 29 PRESSURE PLANE ISOMETRIC VIEW. 46
FIG. 30 VELOCITY PLANE ISOMETRIC VIEW. 46
FIG. 31 AIRFLOW SYSTEM CURVE. 47
FIG. 32 BEST INFERRED CURVE VS CFD SYSTEM CURVE. 48
FIG. 33 PWM TESTING. 49
FIG. 34 FAN IMPLEMENTATION. 49
FIG. 35 LASER/MOUNT ASSEMBLY. 52
FIG. 36 CAD VIEW OF CAVITY. 52
FIG. 37 CAD VIEW OF FLOW STRAIGHTENER INSTALLATION. 53
FIG. 38 CAD VIEW OF INLET ATTACHMENT TO CAVITY. 53

FIG. 39 FRONT VIEW OF HMP60 MOUNT. 54
FIG. 40 CAD VIEW OF EXHAUST ATTACHMENT TO CAVITY. 54
FIG. 41 CAD VIEW OF EXHAUST ATTACHMENT TO CAVITY. 55
FIG. 42 CAD VIEW OF FRONT/REAR BULKHEAD ATTACHMENT TO CAVITY. 55
FIG. 43 FRONT BULKHEAD INSTALLATION GUIDE. 56
FIG. 44 SEALING AND VIBRATION CONTROL GUIDE. 56
FIG. 45 CAD VIEW OF HMP60 SENSOR INSTALLATION. 57
FIG. 46 CAD VIEW OF PAYLOAD ROD STRUCTURE INSTALLATION. 60
FIG. 47 INFILL TENSILE, COMPRESSION, AND THICKNESS TESTS. 64
FIG. 48 SCREW HOLE TESTS. 65
FIG. 49 HEAT SET INSERTS TESTS. 65
FIG. 50 FIT WITH LASER CAVITY, BULKHEAD ENCLOSURE FIT. 66
FIG. 51 FULL ASSEMBLY IMAGES. 67

List of Tables

TABLE 1. SCIENCE TRACEABILITY MATRIX **23**
TABLE 2. SYSTEM REQUIREMENTS TABLE **24**
TABLE. 3 COST BUDGET BREAKDOWN **68**
TABLE. 4 MASS BUDGET BREAKDOWN **69**
TABLE. 5 VOLUME BUDGET BREAKDOWN **70**
TABLE 6. POWER BUDGET BREAKDOWN **71**
TABLE 7. RISK ASSESSMENT **73**
TABLE 8. ELECTRICAL TESTING RISK RETIREMENTS **74**
TABLE 9. MECHANICAL TESTING RISK RETIREMENTS **76**

List of Acronyms/Abbreviations

ADCS	Attitude Determination & Control System
ADC	Analog-to-Digital Converter
ASCII	American Standard Code for Information Interchange
AVR	Microcontroller architecture
BLE	Bluetooth Low Energy
CAD	Computer-Aided Design
CDR	Critical Design Review
CFD	Computational Fluid Dynamics
CG	Center of Gravity
ConOps	Concept of Operations
CSV	Comma-Separated Values
DB9	9-pin D-sub electrical connector
FLAIR	Flying Lab for Airborne Isotope Research
FR-4	Flame Retardant 4
FT	Free Troposphere
GND	Ground
GPIO	General-Purpose Input/Output
GUI	Graphical User Interface
HMP60	Vaisala HMP60 Temperature and Humidity Sensor
INSTAAR	Institute of Arctic and Alpine Research
I/O	Input/Output
ITLL	Integrated Teaching and Learning Laboratory
LM317	Linear Medium 317
M8	Metric 8 Connector
ML	Machine Learning

MoFlowS	Mobile Flow System
ORR	Operational Readiness Review
PCB	Printed Circuit Board
PBL	Planetary Boundary Layer
PDR	Preliminary Design Review
PLA	Polylactic Acid
PWM	Pulse Width Modulation
PyQt5	Python GUI Library
QTimer	PyQt Timer Event Object
R	Resistor
RF	Radio Frequency
RX	Receiving Pin
SQL	Structured Query Language
SRL	Surface Roughness Layer
STM	Science Traceability Matrix
S3	Black Swift S3 drone platform
TPU	Thermoplastic Polyurethane
TX	Transmitting Pin
UART	Universal Asynchronous Receiver/Transmitter
UAV	Unmanned Aerial Vehicle
UTM	Universal Testing Machine
USB	Universal Serial Bus
VIN	Input Voltage
VR	Voltage Regulator
WBS	Work Breakdown Structure
Wi-Fi	Wireless Fidelity

List of Symbols

A	Area
$Amps$	Ampere
C_f	Correction Factor
c	Specific Heat Capacity of Liquid Water
c_d	Specific Heat Capacity of Dry Air at Constant Pressure
H	Relative Humidity
I	Current
k	Ram Pressure Coefficient
L_v	Latent Heat of Vaporization of Water
m	Meters
mm^2	Millimeters Squared
P	Power
PWM	Pulse Width Modulation Duty Cycle
p_o	Reference Pressure
p	Pressure
Q	Volumetric Flow Rate
$q\infty$	Dynamic Pressure
R_e	Reynolds Number
R_d	Specific Gas Constant of Dry Air
R_v	Specific Gas Constant of Water Vapor
R	Rotor Radius
r_t	Total Water Mixing Ratio
r_v	Water Vapor Mixing Ratio
r	Hub Radius
s	Seconds

T	Temperature
V	Velocity
V	Volts
W	Watts
dt	Time step (Difference in time)
mA	Milliampere
mV	Millivolts
t	Time
z	Altitude
°	Angle
$℃$	Celsius
ρ	Pressure
Ω	Ohms
ΔP	Pressure Change
ΔT	Change in Temperature
ΔV	Change in Voltage
Δz	Change in Altitude
λ	Wavelength
μ	Mean Value
RH	Percent Relative Humidity

Chapter 1: Mission Overview and ORR Objectives

1.1 Executive Summary

The Flying Laboratory for Airborne Isotope Research (FLAIR) project is a graduate-level aerospace engineering effort conducted within the ASEN 5018/6028 Graduate Projects at the University of Colorado Boulder. The objective of FLAIR is to design, construct, integrate, and validate an unmanned aerial vehicle (UAV) mounted atmospheric isotope sensing payload, capable of measuring $\delta^{18}O$ in atmospheric water vapor aboard a Black Swift S3 UAV. This Operational Readiness Report (ORR) documents the results of payload construction, subsystem testing, and integrated system validation activities and establishes that the system is prepared to execute the mission defined at Critical Design Review (CDR).

For this project, the main research question focuses on whether a UAV mounted, fixed multi-pass Herriott cell laser measurement system can accurately characterize $\delta^{18}O$ in atmospheric water vapor to improve understanding of moisture transport and mixing processes in the lower troposphere. Ground based and airborne laser spectrometers have already demonstrated that compact cavity enhanced systems can resolve $\delta^{18}O$ at precisions on the order of tenths of a per mil at atmospheric concentrations, using techniques such as optical feedback cavity enhanced absorption spectroscopy, tunable diode laser absorption spectroscopy, and off-axis integrated cavity output spectroscopy [1]. These instruments have demonstrated consistent relationships between $\delta^{18}O$ and thermodynamic conditions associated with condensation and mixing between air masses [2]. Building on this work, the FLAIR hypothesis is that if $\delta^{18}O$ in atmospheric vapor is measured in flight using a compact multi-pass Herriott cell and tunable laser absorption system, then the retrieved $\delta^{18}O$ ratio will correlate with local humidity, temperature, and altitude in ways that can be interpreted in terms of moisture source regions and phase change histories. This would allow FLAIR to observe contrasts between boundary layer and free tropospheric air, identify signatures of evapotranspiration or recent condensation, and evaluate specific phase change processes at a given time and location, consistent with patterns documented in previous field campaigns [2].

The aircraft's experimental system integrates a Vaisala HMP60 humidity and temperature sensor, a multi-pass Herriott cell laser optical cavity, and a customized airflow channel mounted within the payload cavity of a Black Swift S3 UAV. During flight, ambient air is actively pulled through the front end inlet, conditioned through a flow straightener, and directed through the Herriott cell, where multiple internal reflections effectively increase the optical path length and enhance absorption sensitivity, allowing for more precise $\delta^{18}O$ detection. Environmental and system parameters, such as temperature, relative humidity (RH), revolutions per minute (RPM), optical lock state, and cell pressure are all recorded locally and transmitted via S3 avionics and downlink to the ground station. A custom Python/PyQt5 graphical user interface (GUI) provides

real time visualization of humidity, temperature, $\delta^{18}O$ stability, and payload health, while also allowing operators to monitor and adjust airflow or verify optical alignment mid flight.

The payload design maintains stable optical alignment, consistent flow conditions, and thermal control within the Herriott cell. A high static pressure fan and flow straightener supply approximately 2-9.5 CFM of forced flow into the cell, with RPM adjusted dynamically based on mass flow rate feedback to maintain steady mass flow and minimize turbulence. The Arduino Nano microcontroller handles sensor acquisition and component control via universal asynchronous receiver/transmitter (UART) communication, ensuring compatibility with existing S3 avionics. System electronics were verified through analog to digital converter (ADC) resolution testing, which confirmed sufficient voltage supply (~5 mV) for stable $\delta^{18}O$ retrieval. Structurally, lightweight Polylactic Acid (PLA) 3D printed oval bulkheads reduce mass while keeping the payload aligned with the S3's internal mounting geometry.

Structural verification testing was conducted to validate compliance with mass, strength, and shock requirements, and showed the ultimate loading was recorded between 735 N and 990 N. Tensile and compression testing of 3D printed PLA dog bone specimens with varying infill percentages of 30% to 70% were performed using a Universal Testing Machine (UTM) in ITLL Lab 2B36, as documented in the FLAIR tensile and compression test form. Results demonstrated that mid range infill percentages provided favorable strength-to-weight performance compared to higher infill configurations, which exhibited diminishing strength-to-mass returns and more brittle failure behavior. Based on these findings, structural components were manufactured using an optimized infill percentage of 50%. Requirement SR12, which mandates minimization of payload mass while maintaining structural integrity, is therefore verified.

Additional tensile testing of 3D printed specimens with varying thicknesses (6 mm, 8 mm, and 10 mm) was conducted to evaluate load-bearing performance relative to material usage. The 10 mm specimen exhibited an ultimate load of approximately 990 N, the 8 mm specimen approximately 900 N, and the 6 mm specimen approximately 735 N. Load capacity increased sub-linearly with thickness, confirming that reduced thickness components provide substantial mass savings with acceptable structural margin. Based on these results, final structural thicknesses were selected at 6 [mm]. Collectively, structural testing verifies compliance with mass and load requirements with margin.

Adhesive bonding between carbon fiber rods and PLA bulkheads was evaluated under representative assembly and shock conditions as documented in the Adhesive 1 test form. JB Weld Plastic Bonder was applied following surface cleaning and cured for 24 hours prior to shock testing. Hammer induced shock loading at room temperature produced no separation or structural damage. Subsequent cold environment exposure testing was conducted in the Sustainability, Energy and Environment Complex (SEEC) cold room at -24°C to evaluate adhesive performance under anticipated environmental conditions. Bond integrity was maintained under these conditions.

Requirement SR6, which addresses adhesive survivability in colder environments, is assessed as met for the tested environmental extreme of -24°C with residual verification to occur during flight operations.

Electrical and avionics integration testing was conducted in person at Black Swift Technologies on February 16, 2026. This session included direct interface verification between the FLAIR payload electronics and Black Swift Technologies' S3 avionics architecture. Electrical validation activities included confirmation of power supply compatibility at 10, 12, and 19.2-25.2 V, verification of UART communication stability at 9,600 (though might be boosted to 38400 for NeoSWSerial or 57600 for AltSoftSerial libraries) and 115,200 baud, confirmation of sensor acquisition rates of ≤1 Hz, and verification of telemetry transmission through the S3 downlink system. Mechanical component fitting within the S3 payload bay confirmed proper alignment, mounting clearance, and structural attachment integrity. No interference with vehicle systems was observed, and all electrical interfaces operated within expected tolerances. This integration milestone reduced interface risk and verified compatibility with the operational vehicle environment.

Integrated ground testing of the complete payload system was conducted on March 27, 2026. These tests verified fan RPM control authority for requirement SR14, environmental sensor functionality for requirement SR24, GUI telemetry visualization for requirement SR17, synchronized data logging for requirement SR17, and system startup and shutdown procedures for requirement SR7. Optical alignment stability within the Herriott cell was verified under vibration and airflow conditions representative of expected flight loads for requirement SR8.

Position and accuracy validation of the tunable laser absorption system will be conducted during an approximately 15 minute dedicated flight segment within the full mission profile. During this period, steady altitude flight will be maintained to evaluate short term $\delta^{18}O$ measurement stability under operational vibration, thermal, and airflow conditions. Laser lock state, absorption peak stability, baseline drift, and signal-to-noise ratio will be recorded continuously. Allan deviation analysis will be performed on the $\delta^{18}O$ retrieval to quantify precision as a function of integration time, with particular emphasis on 1 to 60 seconds averaging intervals relevant to vertical profiling.

Measurement accuracy will be assessed by comparing retrieved $\delta^{18}O$ trends against expected humidity altitude correlations and by evaluating consistency between ascent and descent profiles. Any systematic offset, drift rate, or vibration induced spectral broadening will be documented. This flight segment will validate that the Herriott cell optical path length, laser tuning stability, and absorption feature retrieval remain within operational tolerance during airborne conditions. Final precision and accuracy metrics (1% precision at 10 minute averaging time intervals and a drift rate of 0.1%/hr) will be inserted following post-flight data reduction.

The mission to be executed consists of a vertical atmospheric profiling flight within the FAA approved COA operations area under Class E and G airspace jurisdiction of Denver ARTCC (ZDV) and Minneapolis ARTCC (ZMP), as defined in COA 2023-WSA-13880. The primary objective of this mission is to obtain in-situ $\delta^{18}O$ measurements during ascent and descent to characterize vertical structure in atmospheric water vapor isotopic composition. The planned maximum altitude is approximately 1,500 meters above ground level (AGL), remaining within the authorized ceiling defined by the COA. If ambient temperatures are sufficiently cold to permit free tropospheric access, the profile will extend through the lower free troposphere; otherwise, measurements will focus on detailed characterization of the planetary boundary layer. Total flight duration is expected to be approximately 60 minutes, allowing for ascent, level off sampling, and controlled descent profiling. Launch and recovery will occur within the defined COA polygon in eastern Colorado, consistent with all ATC coordination and NOTAM requirements specified in the authorization.

The payload has been validated to operate within the environmental conditions expected for this mission. Thermal testing confirmed functionality across a temperature range of -24°C to 60°C during ground validation, satisfying SR5 (thermal testing) and SR9 (functional operation within -30°C to 45°C) with margin. Cold environment adhesive survivability was additionally verified at -24°C, satisfying SR6. Electrical subsystem functionality at low temperature was confirmed down to -20°C, satisfying SR20. All subsystems with defined environmental requirements have demonstrated functionality within these bounds, as documented in the verification matrix.

All construction activities are complete. Subsystem level verification testing has been performed, documented, and can be found in Chapters 3, 4 and 5. Based on the results presented in this ORR, the FLAIR payload is assessed as operationally ready to perform the mission defined at CDR. No open verification items remain that preclude flight, pending final pre-flight inspection and range approval.

1.2 Mission Objectives and Success Criteria

The primary mission objective of the FLAIR project is to successfully collect in-situ measurements of $\delta^{18}O$ in atmospheric water vapor using a UAV mounted sensing payload. This objective directly supports the broader scientific goal of improving understanding of moisture transport, phase change processes, and atmospheric mixing within the planetary boundary layer and lower free troposphere.

To achieve this objective, several measurable success criteria have been defined. First, the payload must maintain stable optical alignment and continuous laser lock throughout the primary data collection segment of the flight. Second, the system must successfully acquire, and log synchronized environmental and isotope data, including temperature, relative humidity, and $\delta^{18}O$ measurements, all at the required sampling rate without data loss. Third, the airflow system must

maintain a consistent and controlled flow rate through the Herriott cell to ensure repeatable measurement conditions.

From an operational perspective, success also requires that the payload integrates seamlessly with the Black Swift S3 avionics system, including stable power delivery, reliable UART communication, and successful real time telemetry downlink to the ground station GUI. Finally, the mission will be considered successful if post-flight analysis demonstrates physically reasonable trends in $\delta^{18}O$ with respect to altitude and humidity, indicating that the system is capturing meaningful atmospheric structure.

1.3 Mission Schedules

Figure 1 shows the PDR schedule. This schedule displays the flow of work for the PDR timeline and the task dependencies experienced. It represents the first major development milestone and establishes the baseline system concept. Early tasks, such as work breakdown structure (WBS) updates, draft budgeting, and component trade studies, create the foundation needed to make informed design decisions. These tasks directly support the mission scope by ensuring that the scientific goals (accurate $\delta^{18}O$ measurements, stable optical alignment, and appropriate airflow) can be met with feasible components. The subsystem design tasks (CAD modeling, airflow assumptions, and initial electrical requirements) were carried out in parallel with requirements drafting. This overlap was necessary because the Herriott cell optical design and airflow parameters evolve together. After the design work stabilized, the team consolidated system level requirements and produced the PDR technical report and slide deck. The PDR concluded successfully with a formal presentation on October 8, 2025.

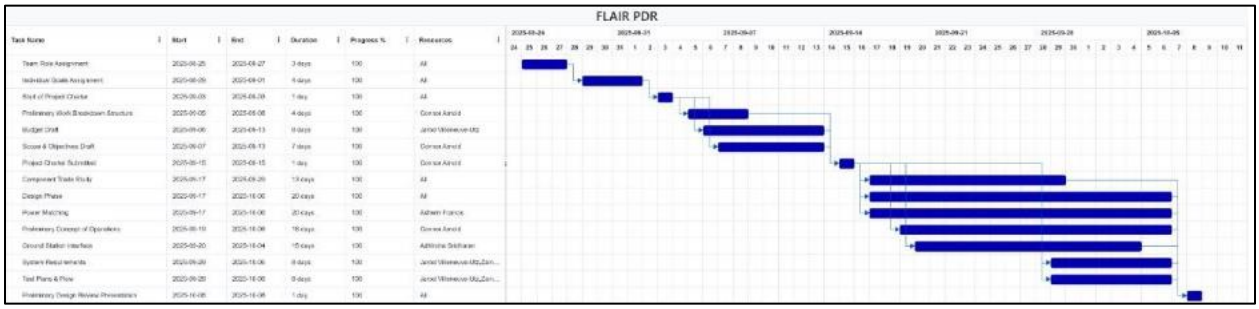


Fig. 1 PDR schedule.

Figure 2 shows the CDR schedule. This schedule displays the flow of work for the CDR timeline and the task dependencies experienced. It focuses on advancing the system from a conceptually validated design to one that is fully ready for fabrication and integration. During this phase, the project emphasizes component level verification, refinement of the CAD models, completion of airflow modeling using both quasi-1D analysis and CFD, and finalization of the electrical schematic and communication framework. Updated mass and power budgets are also developed to ensure that the payload remains within the S3's allowable limits. These tasks support the mission scope by demonstrating that the final configuration can achieve accurate airborne

isotope measurements under realistic flight conditions. Throughout the CDR timeline, the team finalizes component selections for the fan, inlet geometry, electronics layout, and Herriott cell integration, while the PCB design and embedded systems communication plan are completed in parallel. Airflow analyses verify consistent mass flow and optical cavity stability, and structural studies confirm that the bulkheads, ducts, and mounts can withstand expected flight loads. The WBS and budget are updated to reflect the final subsystem decisions, and the schedule culminates in the completion of the CDR report and presentation. The CDR concludes with an internal review demonstrating that the design supports safe fabrication, integration, and alignment with the project's scientific objectives.

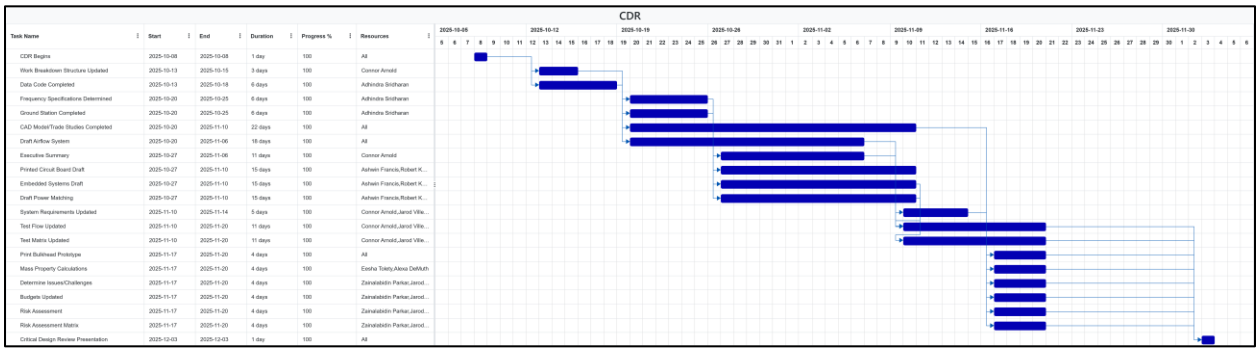


Fig. 2 CDR schedule.

Figure 3 shows the timeline and the task dependencies experienced in the ORR schedule. This schedule represents the final phase of the project timeline and focuses on system integration, validation, and operational readiness for flight. The ORR schedule builds directly on the work completed during the CDR phase by transitioning from design verification to full system validation. Early tasks in this phase include final fabrication of structural components, completion of electrical assembly, and integration of all subsystems into the flight ready payload configuration. These efforts ensure that the physical system matches the finalized design baseline.

Following assembly, the schedule emphasizes subsystem and integrated testing. This includes electrical validation, airflow performance testing, structural verification, and optical alignment checks. Ground testing activities, such as full system power on tests, GUI validation, and telemetry verification, are conducted to confirm that all subsystems operate together as intended. Additionally, testing and validation tasks occur in parallel (when possible) to reduce schedule risk while maintaining sufficient time for troubleshooting and iteration. The final portion of the ORR schedule focuses on documentation, including completion of the ORR technical report, verification matrix updates, and preparation for the ORR presentation. It concludes with a formal readiness assessment, where the team demonstrates that all requirements have been verified and that the system is prepared for safe and successful flight operations.

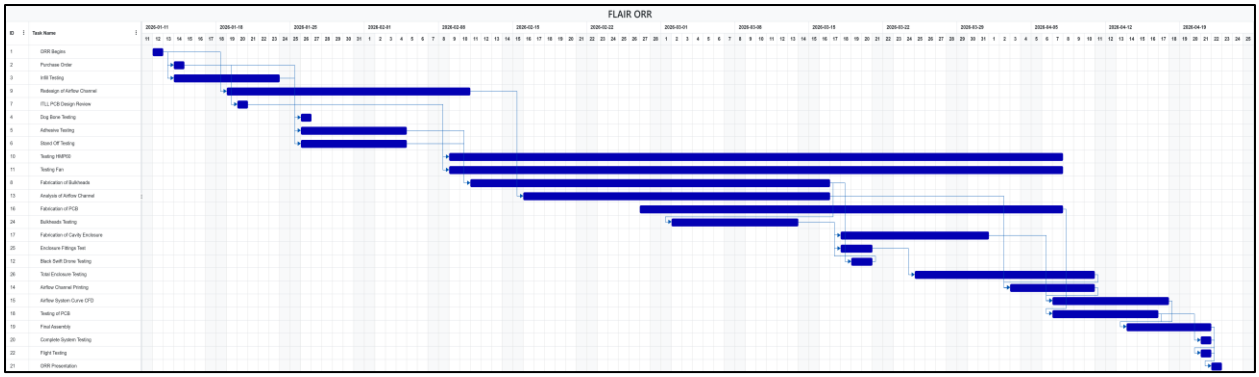


Fig. 3 ORR schedule.

1.4 Work Breakdown Structure

The WBS in Figure 4 organizes the FLAIR project into logical task blocks, each directly aligned with the mission scope of designing, fabricating, and validating the UAV mounted oxygen isotope instrument. This diagram corresponds specifically to sections included throughout this document and provides a baseline reference for subsystem work responsibilities and how they relate to overall system operations.

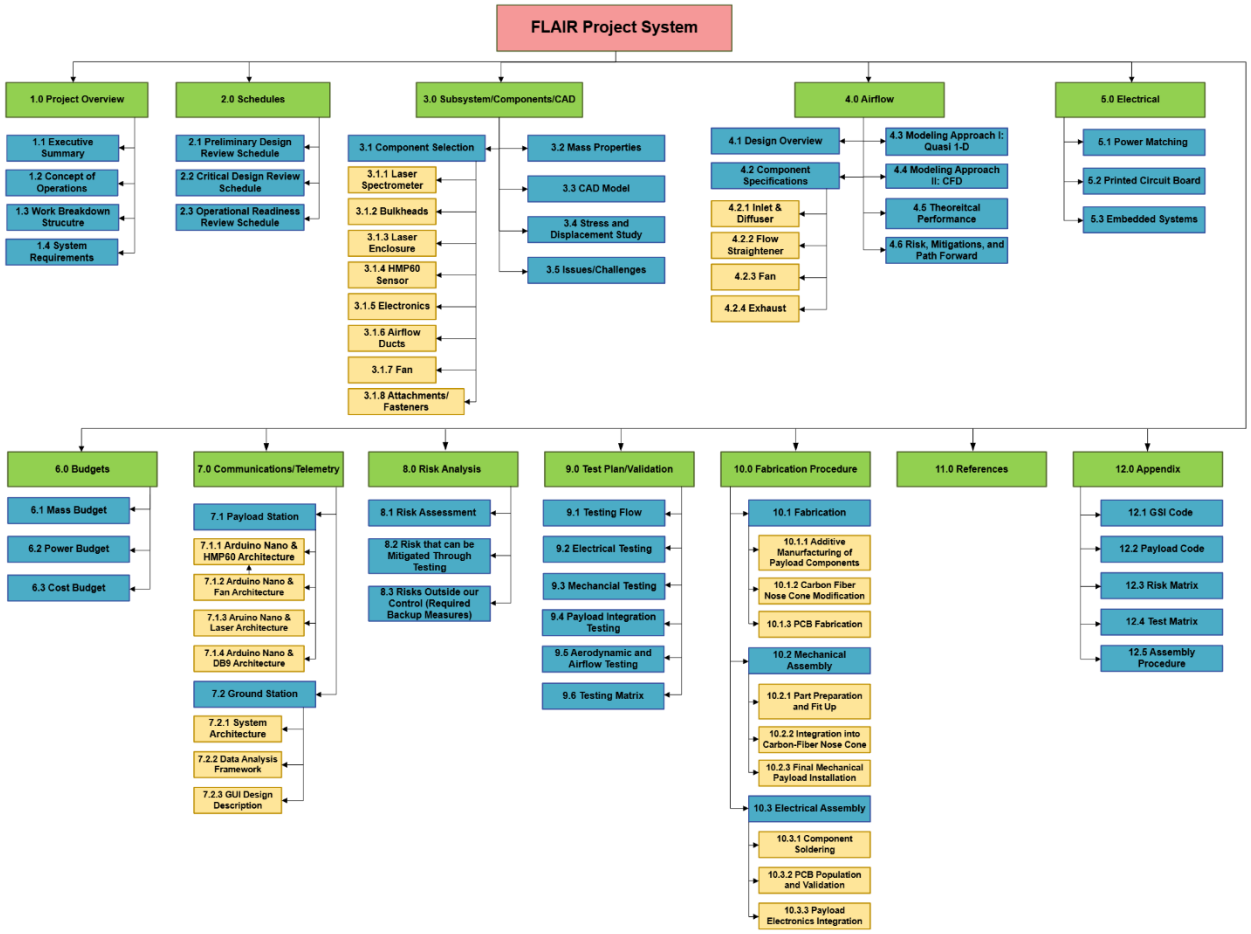


Fig. 4 FLAIR work breakdown structure.

1.6 Flight Readiness Criteria

Flight readiness for the FLAIR mission is determined through a combination of verified technical performance, successful integration, and demonstrated operational capability.

First, all system requirements must be verified through analysis, testing, or inspection, as documented in the requirements verification matrix. This includes confirmation of structural integrity, electrical functionality, airflow performance, and optical system stability. Second, the payload must demonstrate successful end-to-end operation during ground testing. This includes system startup, stable sensor data acquisition, real time telemetry transmission, GUI visualization, and proper system shutdown. No critical failures or unresolved issues may be present. Third, integration with the Black Swift S3 platform must be fully validated. This includes secure mechanical mounting, proper electrical interfacing, and verified communication with the aircraft avionics system. The payload must operate without interfering with aircraft systems.

Environmental readiness is also required. The system must demonstrate functionality within the expected temperature, pressure, and humidity ranges for the planned mission profile. Finally, all operational procedures, including pre-flight checks, mission execution steps, and post-

flight data handling, must be documented and reviewed. The team must be prepared to execute both nominal and contingency operations. Once these criteria are satisfied and reviewed, the system is considered ready for flight pending final range approval and pre-flight inspection.

Chapter 2: Concept of Operations

The Concept of Operations (ConOps) outlines how the FLAIR system will execute its mission objective of collecting in-situ $\delta^{18}O$ water-vapor isotope measurements from a UAV platform. This operational flow directly supports the mission scope described in the Executive Summary, where the project goal is to evaluate whether a compact multi pass Herriott cell laser system, integrated into a Black Swift S3 UAV, can reliably gather atmospheric isotope and environmental data during flight.

The ConOps reflects this mission intent by organizing the entire experiment into a sequence of controlled stages that ensure the payload experiences stable airflow, verified instrument alignment, and predictable environmental conditions prior to, during, and immediately after data collection. The staged timeline also mirrors the system level requirements associated with maintaining optical lock, achieving proper flow rates, and continuously monitoring humidity, temperature, and $\delta^{18}O$ stability throughout the sampling window.

Stage 1 establishes the operational baseline needed for reliable airborne isotope measurements. During this phase, the payload is powered on, the Herriott cell laser system warms up, and the Arduino based electronics begin continuous monitoring of temperature, humidity, RPM, and optical lock status. During Stage 2, the UAV transitions into a stable cruise segment where airflow entering the inlet and passing through the Herriott cell meets the designed scientific conditions. Stage 3 is the mission's scientific purpose: data acquisition loops cycle at high frequency while the ground station, via the GUI, monitors telemetry such as humidity, temperature, RPM, optical stability, and isotope ratios. Finally, Stage 4 ensures orderly shutdown after the aircraft lands, providing a controlled end to the experiment and the required conditions for data retrieval and initial verification.

ConOps is displayed in Figure 5 and the Expanded ConOps is displayed in Figure 6. For nearly the entire ConOps, the MET format used is HH:MM:SS to indicate the expected mission timeline in hours, minutes, and seconds. The time stamps have all been added to the substages of the expanded ConOps as well for an even further breakdown of in-flight operations.

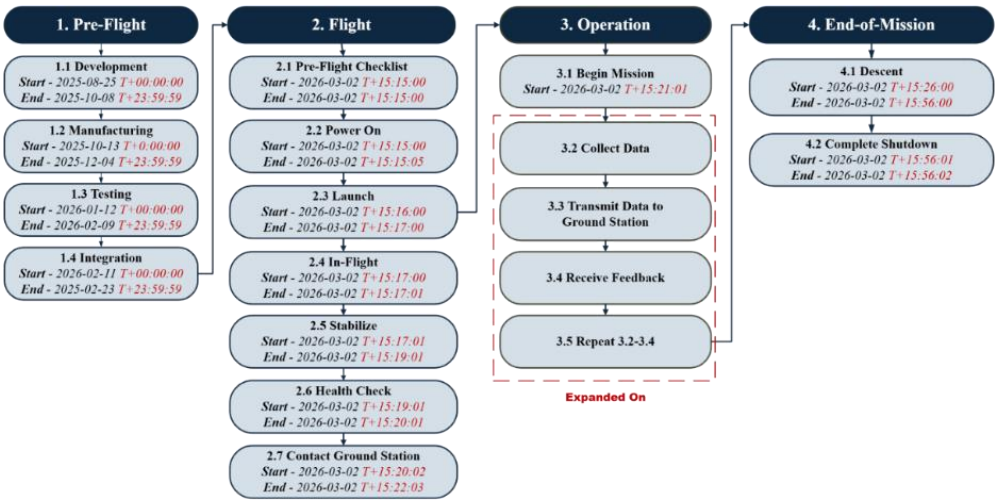


Fig. 5 Concept of operations.

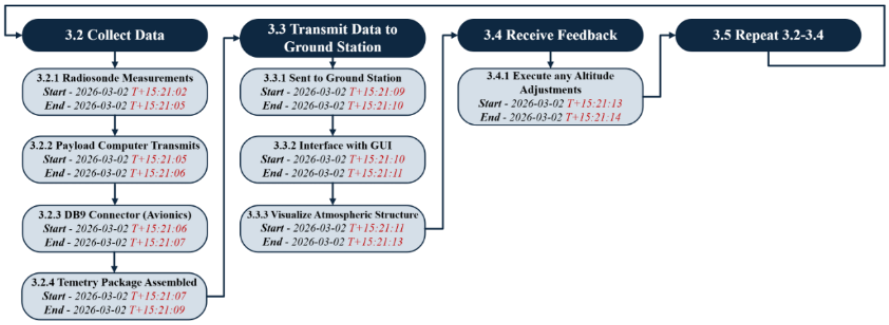


Fig. 6 Expanded concept of operations.

2.1 Operational Mission Profile

The FLAIR operational mission profile consists of a vertically resolved atmospheric sampling flight designed to capture variations in $\delta^{18}O$ as a function of altitude. The mission begins with system initialization on the ground (Stage 1), during which the payload is powered on, all subsystems are verified, and the pre-flight checklist (2.1) is completed.

Following clearance, the UAV lifts off and ascends to a target altitude of approximately 1,500 meters AGL (Stage 2.3), during which the payload continuously collects environmental and isotope data to characterize vertical gradients in humidity and isotopic composition. Once the target altitude is reached, the UAV performs a short level flight segment (Stage 3.1) to capture steady-state atmospheric conditions.

The system then enters its primary data collection loop (Stages 3.2-3.4), during which isotope and environmental data are acquired, transmitted to the ground station GUI, and evaluated in real time to execute any necessary altitude adjustments. The descent phase (Stage 4.1) mirrors the ascent profile, providing a second dataset that can be compared for consistency and used to

identify temporal or spatial variability. The mission concludes with landing and controlled system shutdown (Stage 4.2), followed by data retrieval and initial verification for post flight analysis.

2.2 Pre-Flight Operations

Pre-flight operations focus on preparing both the payload and ground systems for safe and effective mission execution. This begins with mechanical inspection of the payload, ensuring all structural components are securely mounted and free of damage. Electrical checks are performed to verify proper power delivery, sensor functionality, and communication between the payload and the S3 avionics system. The GUI is initialized at the ground station to confirm telemetry reception and real time data visualization. The Herriott cell laser system is powered on and allowed to warm up to achieve stable optical conditions. Fan operation and airflow through the system are verified to ensure proper conditioning of incoming air. A final systems check is conducted to confirm that all subsystems are operating nominally before takeoff clearance is given.

2.3 In-Flight Operations

In-flight operations are divided into controlled phases to ensure stable and reliable data collection. During ascent, the payload continuously records environmental parameters and isotope measurements while operators monitor system health through the GUI. Once stable flight conditions are achieved, the system enters its primary data acquisition phase. During this time, airflow, laser operation, and sensor performance are closely monitored to ensure data quality. The UAV maintains controlled flight conditions to minimize disturbances that could impact measurement stability. Telemetry data is continuously downlinked to the ground station, allowing operators to identify any anomalies in real time. Throughout the flight, system performance metrics such as RPM, temperature, and signal stability are tracked to ensure compliance with operational requirements.

2.4 Post-Flight Operations and Data Handling

Following landing, the payload is powered down in a controlled manner to prevent data corruption and ensure safe shutdown of all components. The system is then disconnected from the aircraft and transported for data retrieval. All recorded data, including environmental measurements, isotope data, and system telemetry, is downloaded and stored in structured formats such as CSV files. Initial data verification is performed to confirm completeness and identify any anomalies. Post-flight analysis includes synchronization of datasets, filtering of noise, and preliminary evaluation of $\delta^{18}O$ trends relative to altitude and humidity. These steps ensure that the collected data is suitable for further scientific analysis.

2.5 System Requirements

General system requirements for a comprehensive mission profile are developed from the scientific justifications of this mission. Table 1 reviews the science traceability matrix (STM), which concisely characterizes the motivation for FLAIR and outlines measurements needed to realize these goals.

Table 1. Science Traceability Matrix

Science Goal	Science Objective	Key Observations	Measurement Requirements	Mirror and Instrument Requirements
Advance understanding of atmospheric water cycle processes and their influence on climate dynamics.	Characterize atmospheric water vapor isotopes ($\delta^{18}O$, δD) using a UAV mounted laser spectrometer to identify moisture sources and phase-change processes.	-Isotopic composition of atmospheric vapor. -Correlated humidity, temperature, and altitude data. -Spatial and temporal variations of $\delta^{18}O$ and δD	- Maintain continuous, stable airflow of 10–22 CFM. - Operate within 12–24 V, <1 A power envelope. - Record ambient temperature and humidity. - Measure altitude and GPS position continuously.	-Laser based spectrometer with active air sampling -Vaisala HMP60 humidity & temperature sensor integrated with UAV avionics.
Improve understanding of moisture transport and isotopic fractionation in the boundary layer.	Quantify relationships between isotope ratios and environmental variables (humidity, temperature, altitude).	-Isotopic gradients with altitude and humidity. -Temporal evolution of isotope ratios during flight profiles. -Correlation of isotopic and environmental parameters.	-Vertical resolution ≤ 10 m -Validation against reference ground data	-High static pressure fan and flow straightener -Arduino Nano control of flow rate via altitude feedback -Real-time GUI monitoring.
Contribute to improved climate and hydrological modeling.	Enhance understanding of lower atmospheric moisture transport and energy exchange.	-Vertical/horizontal isotopic variability. -Boundary layer moisture gradients. -Real time environmental stability during sampling.	-Continuous sampling for full flight duration -Cavity temperature shall remain within operating outside range of −20°C to +45°C	-Thermally isolated, vibration-damped laser cavity -Stable 12–24 V power subsystem .
Demonstrate feasibility of compact UAV based isotope sampling as an alternative to traditional aircraft systems.	Validate performance of UAV mounted isotope payload and operational workflow.	-Stable system operation in flight environment. -Comparison of airborne isotope data with ground reference data. -Power and telemetry integrity.	-Flight endurance ≥ 30 min -Payload mass ≤ 10 kg	-Black Swift S3 UAV platform -Modular payload housing -Telemetry integrated GUI for live operator control.

From these scientific guidelines, overall system requirements and their verification strategies were established for CDR. At this time, due to environmental constraints and timeline limitations, some system requirements have had alterations to meet the scope of what was possible by the team. Table 2 shows the updated system requirements, outlining each subsection requirements, how to verify them, and the method used to do so. The requirements in orange are those that needed modification to their verification strategies.

Table 2. System Requirements Table

SR#	Requirements	Verification Model					Verification	Status/Change
		A	O	D	T	R		
	Presentation Specifications							
SR1	The team shall complete a Preliminary Design Review on 10/08/2025.	x					The team will analyze the timeline of the Project Charter prior to presentation date to ensure completion.	
SR2	The team shall complete a Critical Design Review on 12/03/2025.	x					The team will analyze the PDR project timeline prior to presentation date to ensure completion.	
	Budget							
SR3	The drone payload shall have a material and construction cost not exceeding $2,000 USD			x		x	Mission Manager and System Engineers	
	Environmental Testing Specifications							
SR4	Random vibration testing shall be performed to the levels and duration as defined by the expected atmospheric and weather conditions.				x		The team will select testing facilities that ensure vibration level and duration requirements are met.	Manual shaking/CAD stress testing
SR5	Thermal testing shall be performed to ensure proper operation of components in extreme conditions.				x		The team will select testing facilities that ensure thermal testing requirements are met.	Determined that PLA is can sustain max operational temeratures
SR6	Shock testing shall be performed as defined by the payload manufacturer requriements.				x		The team will select testing facilities that ensure shock level requirements are met.	Manual shaking/CAD stress testing

SR#	Requirements	Verification Model					Verification	Status
		A	O	D	T	R		
	Fabrication and Assembly							
SR7	Interface new components with given CAD file	X					Develop an ongoing assembly with given framework to add on to	
SR8	8G force resistence				X		The team will select testing facilities that ensure shock level requirements are met.	Manual shaking/CAD stress testing
SR9	Function within -30 and 45 deg C	X			X		Test assembly in freezer	Freezer testing no thermal testing
	Structures and Mechancial Systems							
SR10	Fit payload inside of nose cavity (7.972Ø, 24.86L)	X	X				Measure dimensions in completed CAD model, assemble compnents in given nose cone	
SR11	Contain laser system inside of scaffold (16 by 2.5 by 2.5)	X	X				Measure dimensions in completed CAD model, assemble compnents in given scaffolding	
SR12	Mass of payload 5lbs or less	X			X		Measure weight in completed CAD model, weigh payload before assembling	
SR13	CoM of payload 89-114mm from wing leading edge	X		X			Measure weight in completed CAD model, test payload CoM before assembling	
	Air Flow							
SR14	Residence time of air under 1s	X					Model airflow with CFD	
SR15	Accomodate 15 m/s winds	X			X		Model airflow with CFD, use wind tunnel to verify	Could not do wind tunnel verfication. Confident CFD predictions.
SR16	Operate at or beneath 4600m altitude	X					Model airflow with CFD at different atmospheric conditions	Confident in fan power capabilites for verification

SR#	Requirments	Verification Model					Verification	Status
		A	O	D	T	R		
	Communication Specifications							
SR17	Connect to ground station & back within 10 minutes			X	X		Run some code to send a message from the drone to the ground station.	
SR18	Electrical Board Specifications							
SR19	Make sure that internal ADC has 10-bit resolution	X			X		Test in Aerospace Electronics Lab.	
SR20	Capable of working down to -20 C temperatures		X		X		Test in SEEC freezer units.	
SR21	Total system can run on <3A & 19.2V-25.2V of power				X		Test in Aerospace Electronics Lab.	
SR22	6D/2A or 8D pins minimum on board		X				Compare boards having more available pins.	
SR23	Ensure all PCB connections are firm			X	X		Demonstrate connection firmness after assembly	
SR24	Sensor Specifications							
SR25	Sensor accuracies are within documented specifications		X		X		Test in Aerospace Electronics Lab and SEEC freezer units.	

Verification	Meaning
A	Analysis
O	Observation
D	Demonstration
T	Testing
R	Review

	Status
Completed	
Partial	
Changed	

The STM and system requirements tables link the overall scientific motivation to system level requirements, establishing the broader implications of this project while reconciling them with the limitations encountered throughout the team's development process.

Chapter 3: Final System Configuration

3.1 Payload Overview/Updates

Since the Critical Design Review (CDR), several key updates and refinements have been made to the FLAIR system to improve performance, reduce risk, and ensure compatibility with operational constraints.

One of the primary changes includes refinement of the structural design based on testing results. Adjustments to infill percentage and component thickness were implemented to optimize the strength-to-weight ratio while maintaining sufficient structural margin. These updates directly support mass and load requirements while improving overall payload efficiency.

The airflow system also underwent modifications following analysis and testing. Adjustments to inlet geometry, flow straightener configuration, and fan operating conditions were made to achieve more consistent mass flow and reduce turbulence within the Herriott cell. These changes improve measurement stability and support more reliable isotope data acquisition.

Electrical and avionics integration was further refined through in-person testing with Black Swift Technologies. Communication protocols, power interfaces, and sensor acquisition rates were validated and adjusted where necessary to ensure compatibility with the S3 platform. These updates significantly reduced interface risk compared to the CDR baseline.

Additionally, updates were made to the software and GUI to improve real time monitoring capability and data visualization. Enhancements to telemetry handling and user interface responsiveness ensure that operators can effectively monitor payload performance during flight. Overall, these changes represent incremental but critical improvements that increase system robustness and confidence in operational performance.

3.1.1 Mechanical Updates

3.1.1.1 Bulkhead Updates

Through the testing of different infills of 3D printed dog bone shapes (see section 5.2.1), it was determined that an infill of 40% to 60% performed the best under compressive and tension for their weight. Then, the thickness of the bulkheads was tested at 50% infill to determine if further weight could be saved without sacrificing structural integrity of the component that holds up the entire payload. The airflow components that stick out of the payload require the payload nosecone to flex slightly during assembly to fit into place. Therefore, the bulkhead shape had to be shaved off on the curves to allow for extra flexing of the shell during assembly. Also, both the bulkheads on the right side where the laser electronics were mounted had smalls holes for the optical fiber wire.

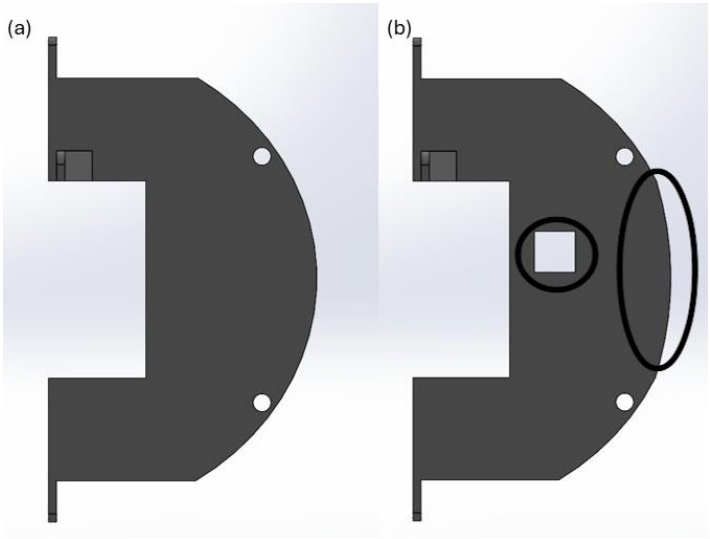


Fig. 7 Bulkhead changes, before (a) and after (b).

3.1.1.2 Electronic Mounting Updates

Heat set inserts were already the method of attachment for the electronics at CDR. But now the places to mount the electronics are raised from the bulkhead to provide better airflow, prevent metal-electronics contact from other screws, and to prevent bulkhead melting.

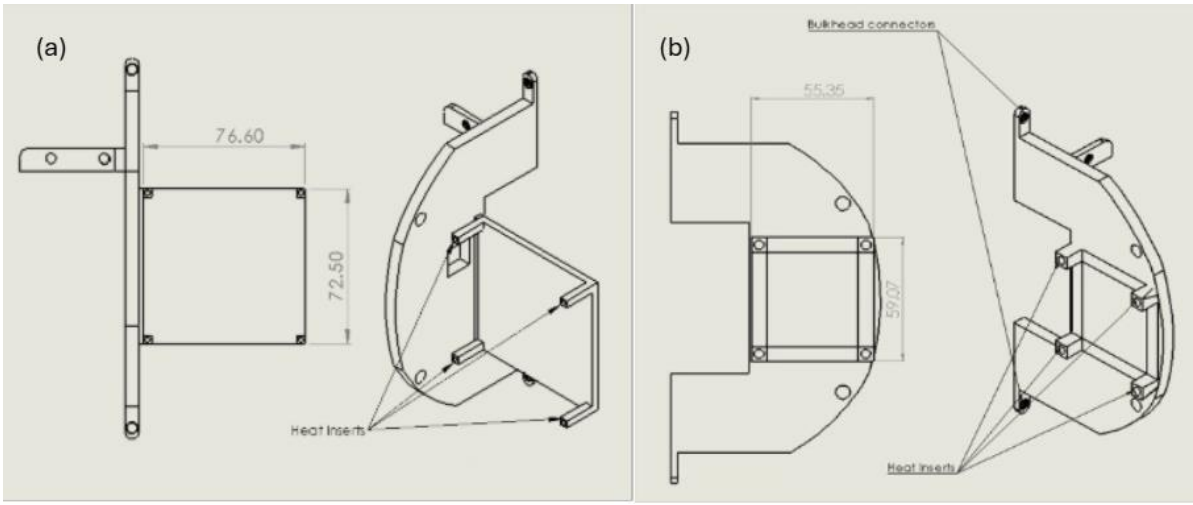


Fig. 8 Diagrams of electronics mounting, laser mounting(a), PCB mounting (b).

3.1.1.3 Enclosure Updates

The exact length of the laser spectrometer was adjusted since CDR. This changed the length of the laser enclosure from 16 to 14.25 in. There were 3 flanges on the bottom of each half of the enclosure where screws would connect the two halves. Once updated, there were only 2 flanges to account for the reduced length of the enclosure.

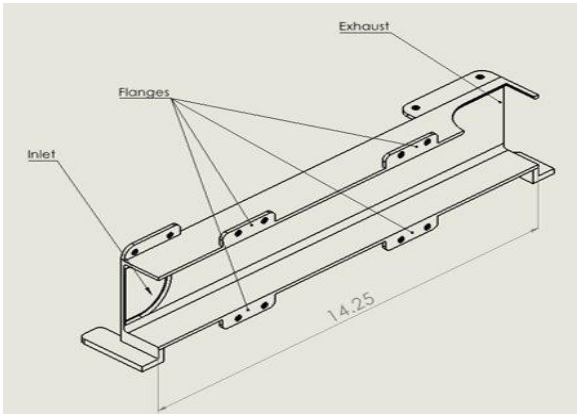


Fig. 9 Modified enclosure CAD diagram.

Dr. Darin Toohey vocalized concerns about torque on the laser spectrometer and the implications of clamps used on each end to attach it to the enclosure. Therefore, once updated, only one end is firmly attached with a clamp, and the other has rubber foam wedged between the

cage plate and enclosure. The rubber foam acts as a shock absorber without creating additional torque on the laser spectrometer.

To prevent air from entering or exiting the laser spectrometer, tape is used on the enclosure. All seams are covered with metallic tape that can sustain cold Arctic temperatures.

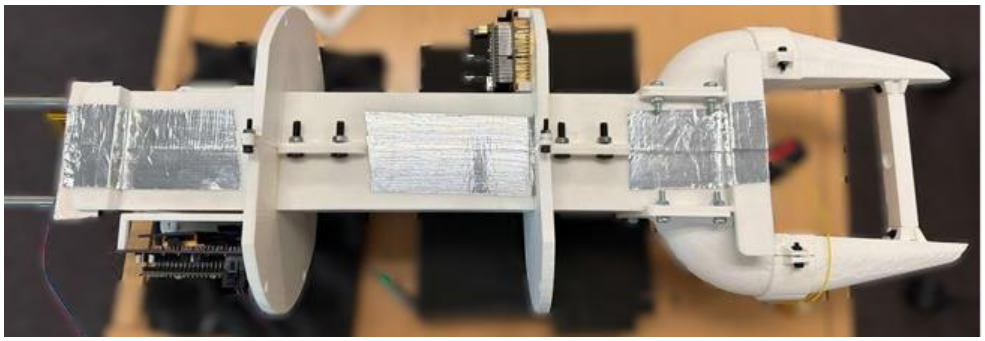


Fig. 10 Enclosure covered with tape.

3.1.1.4 Payload Mounting Updates

The payload needs to stay fixed on the payload rods. The same metallic tape that is used on the enclosure will be used on the rods after the payload is correctly placed, in a way to prevent the payload from moving. The metallic tape is not used in the current assembly, as shown by Figure 11, but it should be used in future flights where low temperatures may compromise other tape's adhesive capabilities. Also, before taping up the payload, look for the position where the center of gravity (CG) is within range of 89-114 mm. If placed incorrectly, there is a high chance of shifting the CG and making it nose-heavy.

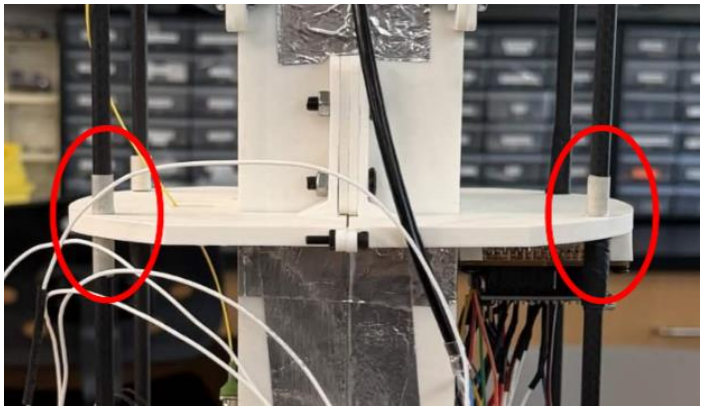


Fig. 11 Payload mounting with tape.

3.1.2 Electrical Updates

3.1.2.1 Integration of Logic Level Shifter

A logic level mismatch existed between subsystems operating at different voltage levels. The Teensy microcontroller operates at 3.3 V, while the Arduino Nano and the laser interface require 5 V logic. Direct interfacing between these components can result in unreliable signal interpretation or potential damage. To address this, a bidirectional logic level shifter was introduced between the 3.3 V and 5 V domains. This ensures proper voltage translation for both directions of communication, improving signal integrity, and system robustness.

3.1.2.2 Replacement of LM317 Linear Regulators with LM7810 Linear Regulator and XL4015 Buck Converter

The earlier design utilized multiple LM317 voltage regulator units in parallel to supply power to the fan and laser. As a linear regulator, the LM317 dissipates excess voltage as heat, leading to low efficiency, particularly under higher current loads. Additionally, parallel operation of linear regulators introduces imbalance and thermal concerns.

This configuration has been replaced with an LM7810 and a buck (step down) converter, which provides simpler designs, higher power conversion efficiency (LM7812 = 40-60%, XL4015 = 96%), increased thermal dissipation, and stable output voltage or current options under varying load conditions. This change significantly improves overall system reliability and enables sustained operation without thermal constraints.

3.2 Mechanical Configuration (As-Built)

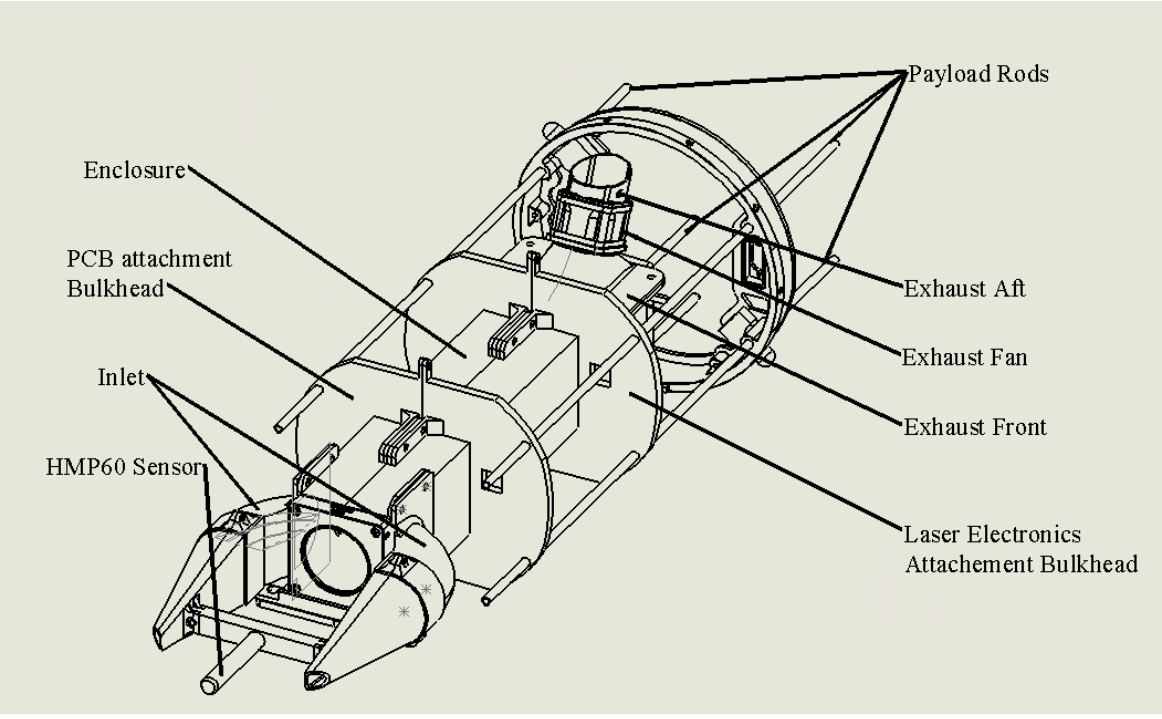


Fig. 12 Full assembly CAD model.

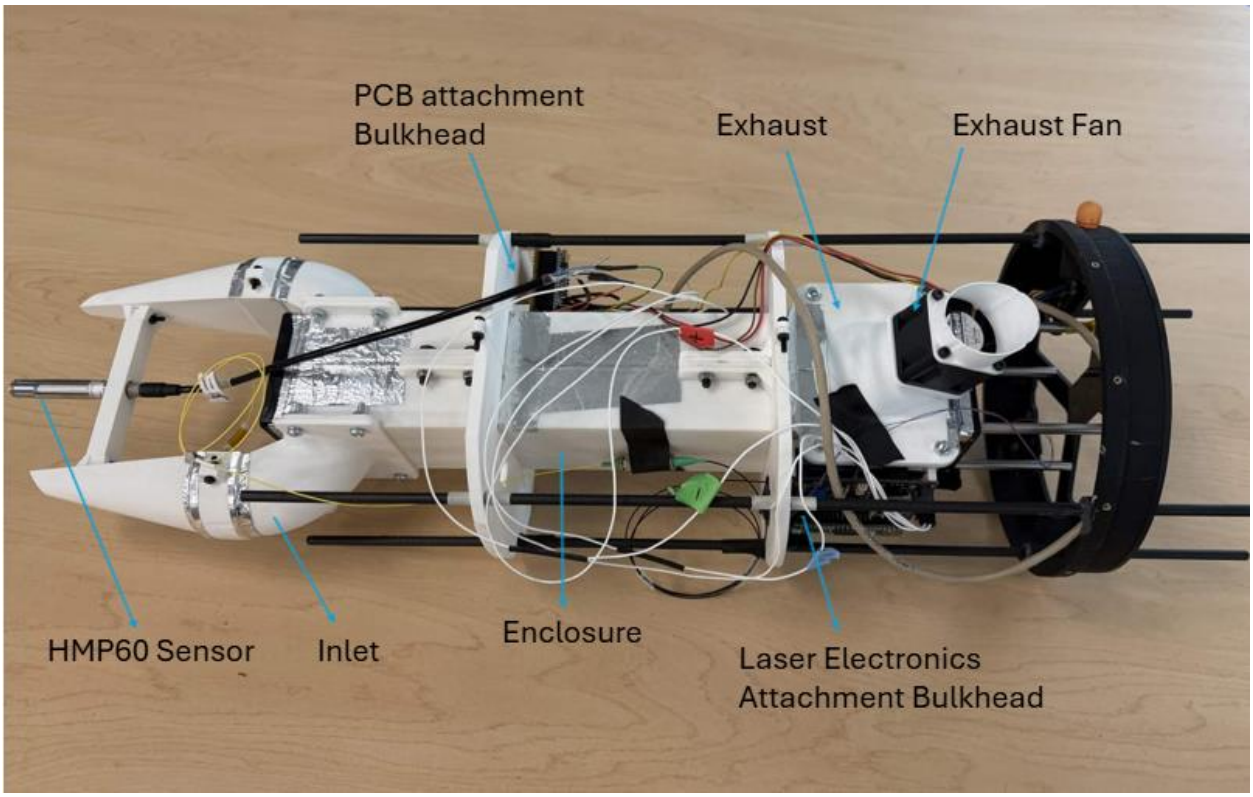


Fig. 13 Assembled payload.

3.3 Electrical Wiring and Connections

The electrical system is organized around a central printed circuit board (PCB), which serves as the primary hub & connector for power distribution, control, and signal routing. All connections are also listed in comments at the top of the Arduino Nano's code and in section 4.3.4.

3.3.1 Power Distribution

Power is supplied through an external connector (DB9 or equivalent) and routed into the PCB. From there, input power enters the regulation stage, the buck converter & LM7810 step down the voltage to required levels, and the regulated power is distributed to: LM7812 = Arduino Nano, Buck Converter = fan and laser, Direct Power = Vaisala HMP60 sensor. This architecture ensures stable and consistent voltage delivery across all subsystems.

3.3.2 Control System (Arduino Nano)

The Arduino Nano serves as the primary controller, responsible for acquiring sensor data, controlling the fan and laser, and managing communication interfaces. Key connections include:

- VIN → regulated power input (7-12V, set to 10V)
- GND → common ground

- Analog pins (A4, A5) → analog sensor inputs
- Digital pins (D0 = built-in RX = D2 [connected via soldered wire on bottom of PCB], D1 = built-in TX = D3 [connected via soldered wire on bottom of PCB], D5, D8, D9) → UART communication, & actuator control (fan)

3.3.3 Sensor Interface (HMP60)

The Vaisala HMP60 sensor measures temperature and relative humidity and is connected via an M8 interface to the cable which came with it. This cable ends in open wires which are then soldered to a connector system which can be plugged into the PCB. Key connections include:

- Power → direct supply from drone's DB9 connection
- Ground → common ground
- Temperature output → Arduino Nano's A5 pin
- Humidity output → Arduino Nano's A4 pin

Analog signal acquisition is used for simplicity and ease of debugging, and stable power supply ensures measurement accuracy.

3.3.4 Actuator Interfaces

Fan:

- Powered directly from buck converter output
- Grounded to common system ground
- Controlled via Arduino Nano's D5 pin set to PWM mode

Laser:

- Robert Kaplan and Dr. Darin Toohey will implement this after the end of the semester
- Powered directly from buck converter output
- Controlled via Arduino Nano's digital pins using UART serial communication
- Signal routed through logic level shifter for the laser

The fan and laser are intentionally not powered through the Arduino to prevent excessive current draw on the controller.

3.3.5 Logic Level Shifter

A bidirectional logic level shifter is implemented between 3.3 V and 5 V subsystems to:

- Convert 3.3 V signals to 5 V

- Convert 5 V signals to 3.3 V

This ensures reliable communication and protects components from voltage incompatibility.

3.3.6 Communication Interface

An external connector (DB9 or equivalent) is used for system interfacing. It carries:

- Power input
- Ground reference
- Communication signals

This enables straightforward integration with external systems and facilitates testing.

3.3.7 Grounding Strategy

Common ground is maintained across all components through the PCB. This:

- Minimizes electrical noise
- Ensures stable signal reference
- Improves overall system reliability

3.3.8 Wire Types and Connections

- **High power lines:** 18 AWG stranded wires (buck converter connection)
- **Low power & signal lines:** 22 AWG solid wires (soldered to remaining connections)
- **Connections:** PCB-mounted female-male header connectors for modularity and ease of testing (for low power & signal lines to plug into)

3.4 Electrical Architecture (v3 = Final)

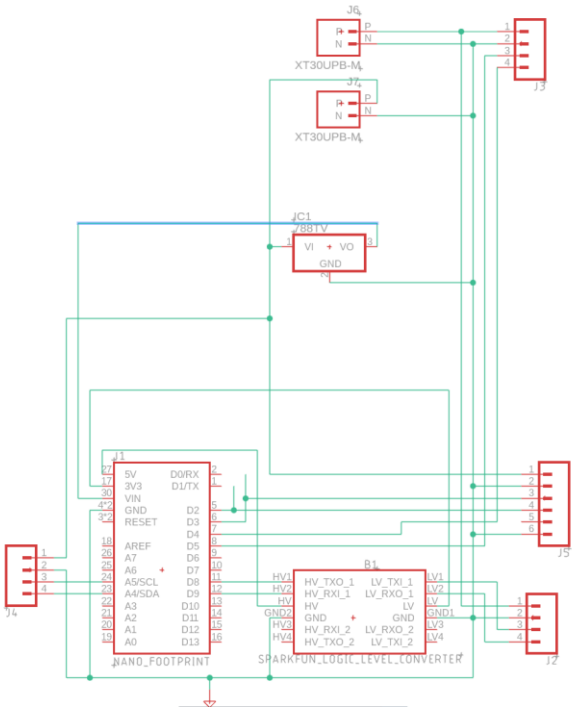


Fig. 14 Final electrical schematic.

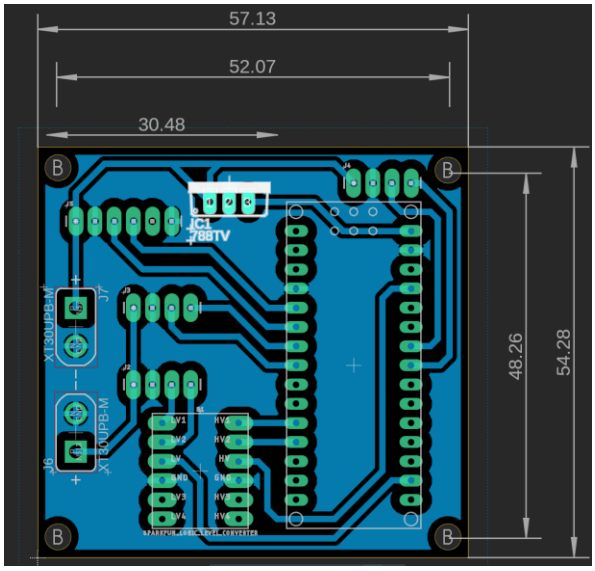


Fig. 15 Final PCB layout.

The final electrical architecture follows a modular and efficient design philosophy.

3.4.1 System Overview

- Single central controller (Arduino Nano)

- Single power input interface (DB9 pins 1 & 2)
- One-stage power conversion (buck converter)
- Distributed subsystem connections via female-male header connectors

3.4.2 Power Architecture

- External power input (DB9 pin 1) → system entry
- Routed through buck converter
- Regulated output supplied to:
    - Arduino Nano
    - Fan
    - Laser
    - HMP60 sensor

This configuration reduces power losses and minimizes thermal load.

3.4.3 Control Architecture

- The Arduino Nano performs:
    - Sensor data acquisition (HMP60)
    - Actuator control (fan and laser)
- A logic level shifter ensures voltage compatibility between subsystems.

3.4.4 Design Characteristics

- **Modular design:** simplifies testing and subsystem isolation
- **Compact layout:** reduces wiring complexity and electrical noise
- **Efficient power conversion:** minimizes heat generation
- **Robust communication:** ensured via level shifting

3.5 Payload Software and GUI Configuration

3.5.1 Payload Software

The payload software is a code written for the Arduino Nano to make it act as the central brain/connector of the payload. It is split into a few primary sections. Initialization/setup, loop, and helper functions.

3.5.1.1 Initialization/Setup

The initialization/setup sections of the payload code include everything from the beginning of the script through the "setup" function.

The initialization section starts with the pin definitions and has details on the physical connections and configuration of the PCB in comments with each related part. This is followed by some global variables & structures copied from Black Swift Technology's gitlab. The final part of the initialization section is the global structures which store the packet information used to send and receive data from the drone.

The "setup" function is always the first function to be called in an Arduino code (after the script is compiled). This is then followed by the infinite "loop" function. The "setup" function starts serial connections, defines pin modes, and sets initial data such as constant values in the packets or the initial fan speed.

3.5.1.2 Loop

The "loop" function is the main body of the code. The loop starts by setting the time that the current loop started and incrementing the packet number by one. It then prints the current loop information if the DB9 mode is set to true (the DBG variable also needs to be set to the number 1 which will print to the computer but won't send serial communication messages). Then it gets & sets (in the packet structure) the current HMP60 data (temperature & relative humidity) using the built-in "analogRead" function, and the current laser data using the custom "readLaser" function. Following this, it then sets a new fan speed if the current one doesn't match the one set in the packet structure. The fan speed is set here to hopefully give as much time as possible before the next data readings for the air to stabilize (though the loop is usually too fast for this to matter). Then it collects any data sent from the DB9 (this portion is not working as of the end of the semester, but Robert Kaplan will work with Jack Elston to get this working after the semester ends). Finally, the loop ends by setting the final time of the current loop and then immediately sending the current packet to the DB9 (drone) to be sent to the ground station.

3.5.1.3 Helper Functions

There are a total of seven helper functions as of the end of the Spring 2026 semester. They are "sendBST", "sendChannel1Status", "sendChannelAck", "setFletcher16", "checkFletcher16", "readDB9", and "readLaser".

The "sendBST" and "sendChannel1Status" functions each send packets. The former sending the actual data packet, and the latter sending a status packet intended to get the current setting of the fan speed (implementation will be changed later). The "sendChannelAck" calls the "sendBST" function. The functions listed in this paragraph were created by Jack Elston at the end of the semester and will be updated as needed after the end of the semester.

The "setFletcher16" and "checkFletcher16" functions are used to generate and check the Fletcher 16-bit checksum for each packet being sent and received. The former was also made by Jack Elston and both will also be updated as needed following the end of the semester.

The "readDB9" and "readLaser" functions are the metaphorical meat of the helper functions as they facilitate the majority of the code's complexity. They are responsible for the reading of information from each device. As of the end of the semester, there is likely an issue with the "readDB9" function since it appears unable to receive information from the ground station. This will be worked on after the end of the semester and will likely combine many of the other functions mentioned earlier into this one. The "readLaser" function is currently working, but a future version will be created by Robert Kaplan and Dr. Darin Toohey to allow the Arduino Nano to also send information to the laser to allow for the laser to modulate itself automatically.

3.5.2 GUI Configuration

The Ground Station Interface is a real time payload monitoring and control system designed to facilitate bi-directional communication between a high-altitude flight payload and a ground operator, as is shown in Figure 16. It integrates live telemetry visualization, packet level diagnostics, remote actuation, and offline data recovery into a unified control deck. The system communicates over a custom Wi-Fi port using a dual channel serial configuration (RX CH0 = 232, TX CH1 = 233).

The top bar of the GUI displays the system identity along with a MODE indicator button in the top-right corner. Immediately below this sits a status bar showing real time stream health metrics like “Test Stream: ____% pkt ____ / _____” and a Serial connection field, confirming active hardware communication. This dual-indicator design enables operators to immediately assess link quality and data throughput at a glance.

The left sidebar houses all primary mission control buttons, organized top-to-bottom in operational sequence:

- Connect WiFi – Establishes wireless uplink to payload
- Connect Serial – Initiates serial port binding
- START mission – Begins active data acquisition
- END mission – Terminates recording session
- LOAD – Loads a data file for processing in offline conditions

A dedicated FAN control [TX CH1 = 233] subsection provides:

- START FAN toggle
- Power Level slider – initial set to 25%, which can be modified to be set to 0 at the start of the flight if needed
- Fine Adjust percentage – initial set at 25%
- Last TX display for confirming last transmitted command

Six real time graphs occupy the central-right area of the GUI, arranged in a 3 x 2 grid:

ROW 1:

- Wind speed vs altitude – It displays wind speed (m/s) across altitude bands, using the wind speed sensor as part of the drone electronics
- Temperature vs Altitude – Plots atmospheric temperatures (°C) vs altitude (m), using the HMP60 radiosonde sensor data, placed at the inlet of the payload

ROW 2:

- Pressure vs altitude – It displays pressure (hPa) across altitude bands, using the pressure sensor as part of the drone electronics
- Relative humidity vs Altitude – Plots humidity (%) vs altitude, using the HMP60 radiosonde sensor data, placed at the inlet of the payload

ROW 3:

- $\delta^{18}O$ Isotope vs altitude – It plots isotope ration (%) across altitude bands, using the laser data, that is recorded after the laser measurements
- UTC and Mission clock – Displays UTC time and Mission Elapsed time in large, high contrast digits for easy readability during operations

The right payload status panel consolidates all numerical telemetry into a compact status readout. The laser metrics (Mixing ratio, background, peak) are scientific outputs from the onboard spectrometer, feeding directly into the isotope calculation. An additional strip chart in the lower right shows Background and Peak laser readings against packet numbers. This allows us to monitor instrument stability and detect any sensor drift/interference in near real time.

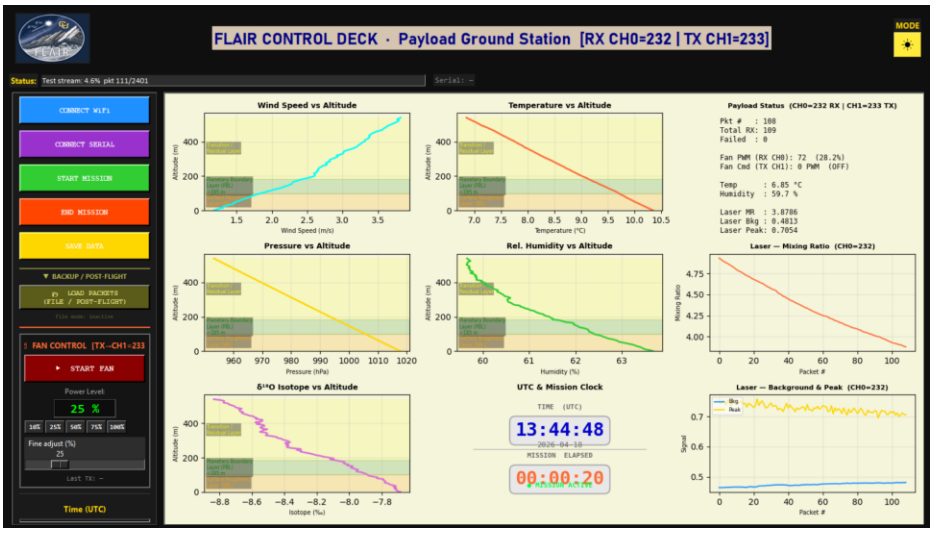


Fig. 16 GUI configuration.

There are a couple of upgrades incorporated into this test-flight version of the GUI, as compared to the version presented in the CDR, held in December 2025.

3.5.3 Upgrade 1: Raw Packet Display (Hexadecimal Diagnostic View)

```
[TEST RAW] pkt#10 42B | 55 24 E8 00 22 00 10 27 00 00 0A 40 34 02 C7 01 10 27 00 00 0A 00 00 00 E8 3A 97 40 B0 C4 EF 3E FE D3 3B 3F E4 2A 00 00 89 B3
[TEST RAW] pkt#11 42B | 55 24 E8 00 22 00 F8 2A 00 00 0B 40 33 02 C7 01 F8 2A 00 00 0B 00 00 00 AE EA 96 40 B4 72 EF 3E 1E F4 3F 3F CC 2E 00 00 CE E2
[TEST RAW] pkt#12 42B | 55 24 E8 00 22 00 E0 2E 00 00 0C 40 33 02 C6 01 E0 2E 00 00 0C 00 00 00 83 70 96 40 DE 11 EF 3E C7 89 40 3F B4 32 00 00 5A 0A
[TEST RAW] pkt#13 42B | 55 24 E8 00 22 00 C8 32 00 00 0D 41 33 02 C5 01 C8 32 00 00 0D 00 00 00 D0 03 96 40 58 BA EE 3E D9 F4 3F 3F 9C 36 00 00 80 4E
[TEST RAW] pkt#14 42B | 55 24 E8 00 22 00 B0 36 00 00 0E 41 32 02 C5 01 B0 36 00 00 0E 00 00 00 A9 86 95 40 86 C4 EE 3E 43 AF 3B 3F 84 3A 00 00 CC C7
[TEST RAW] pkt#15 42B | 55 24 E8 00 22 00 98 3A 00 00 0F 41 32 02 C4 01 98 3A 00 00 0F 00 00 00 31 F7 94 40 37 C5 EE 3E F3 13 3F 3F 6C 3E 00 00 BC 4E
[TEST RAW] pkt#16 42B | 55 24 E8 00 22 00 80 3E 00 00 10 41 32 02 C3 01 80 3E 00 00 10 00 00 00 23 91 94 40 98 09 EF 3E 6F 8F 3D 3F 54 42 00 00 D0 3B
[TEST RAW] pkt#17 42B | 55 24 E8 00 22 00 68 42 00 00 11 41 31 02 C2 01 68 42 00 00 11 00 00 00 A5 19 94 40 EE 5E EF 3E C0 17 3A 3F 3C 46 00 00 DD 8A
[TEST RAW] pkt#18 42B | 55 24 E8 00 22 00 50 46 00 00 12 41 31 02 C1 01 50 46 00 00 12 00 00 00 18 88 93 40 06 4B EF 3E 4D 5C 3F 3F 24 4A 00 00 6C 0C
[TEST RAW] pkt#19 42B | 55 24 E8 00 22 00 38 4A 00 00 13 41 31 02 C1 01 38 4A 00 00 13 00 00 00 02 17 93 40 E4 83 EF 3E 3D 40 3C 3F 0C 4E 00 00 35 32
[TEST RAW] pkt#20 42B | 55 24 E8 00 22 00 20 4E 00 00 14 41 30 02 C0 01 20 4E 00 00 14 00 00 00 FE A9 92 40 EB A5 EF 3E BD 9D 3F 3F F4 51 00 00 F1 8B
```

Fig. 17 Packet displayed in hex stream format.

A dedicated terminal in a separate window from the GUI streams raw packets of data in hexadecimal format as it arrives over the connected Wi-Fi source. Each packet is 42 bytes in length, which is also explained in the Payload code section 3.5.1. The hex display allows the operators and flight testing engineers to verify packet structure, identify byte level corruption if any, confirm number sequencing in the arriving packets, and cross check parsed sensor values against raw content. This becomes particularly critical for high integrity atmospheric missions where sensor data must be traceable with the raw telemetry being sent to the ground. The display serves as a real time debugging layer. If parsed values appear in an anomalous fashion, the operators can inspect the underlying hex stream to determine if the issue is in the sensor, the parsing, or transmission being corrupt. The display hex stream is shown in Figure 17.

3.5.4 Upgrade 2: Fan Speed Control from Ground Station

The FAN Control section of the left panel enables remote PWM (Pulse Width Modulation) actuation of the payload's exhaust fan via the TX channel (CH1 = 233). The operator can adjust power level in coarse increments via the slider and fine tune using the percentage input field. The exhaust fan regulates airflow through the payload cavity, directly influencing the internal temperature and pressure conditions surrounding the laser spectrometer. By controlling the fan speed from the ground:

- The ground operators, testing engineers, and the entire team can slow air throughput to allow the spectrometer more residence time with each air sample.
- Increasing the fan speed flushes the cavity faster, enabling quicker response during rapid altitude transition.
- Real time PWM feedback within 0-255 range confirms that the command was received and executed by the payload, closing the control loop.

This upgrade converts the ground station from just another monitoring dashboard tool to an experimental control platform, enabling in flight optimization of measurement conditions.

3.5.5 Upgrade 3: Offline Post-flight Analysis Mode

The BACKUP / POST FLIGHT section provides a Load Packets function, that uses pre-recorded packet files that are collected during flight or from backup storage. It then processes these packet files through the same parsing, validation, and visualization pipeline as the live stream. Communications dropouts and non-readability are an operational and testing reality, due to atmospheric interference, or range limitations of the Wi-Fi signal. This upgrade ensures that no data is permanently lost due to any of these comms failures. Post-flight, the ground station team loads the raw log file; the system parses each packet, reconstructs the time series telemetry, and regenerates all 6 sensor plots identically to how they would have appeared during real time live tracking telemetry.

To summarize this flight-ready configuration, the GUI represents a well-structured telemetry interface that balances operational simplicity. Its key architectural strengths are:

- Dual channel serial architecture (RX/TX channels separate) that prevents command traffic from interfering with incoming telemetry.
- Layered data visibility – from raw hex bytes up through parsed values and graphical trends.
- Closed loop actuation with feedback, ensuring commanding states of the system are verified.
- Resilient data recovery through offline reprocessing.
- Mission clock integration providing synchronized timestamps for correlation with reference datasets like GPS altitude logs, and radiosonde records.

3.6 Mass Properties and Center of Gravity Verification

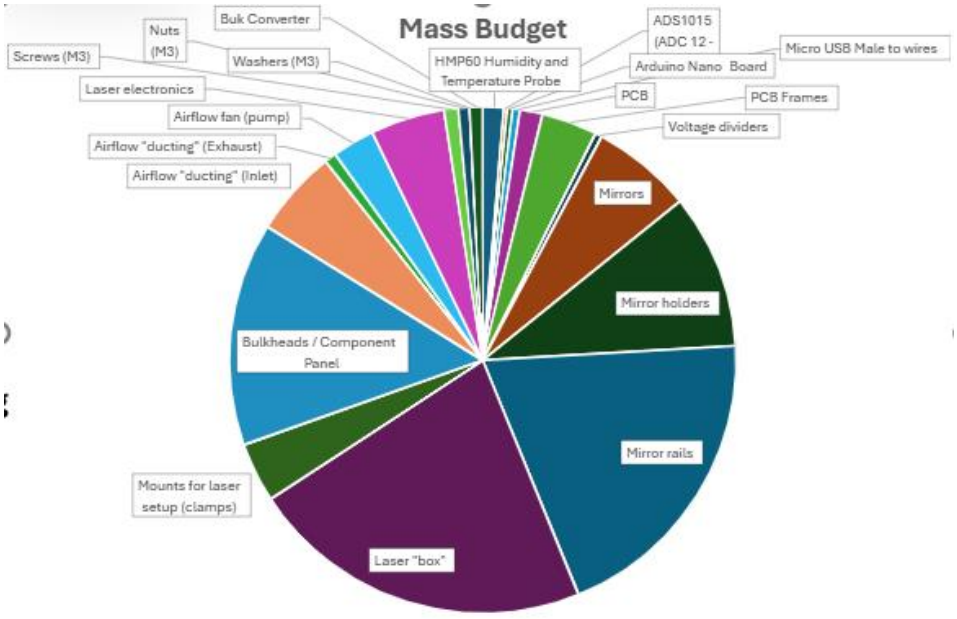


Fig. 18 Mass budget.

In Figure 18, the budget of all of the masses are shown. The upper limit of the mass is 2,267.96 grams, but only 2,106.42 grams was used. It has a margin of 159.54 grams.

Component	Location (mm)	Weight (kg)
Empty drone	-196.5	7.175
Laser Spectrometer (including Enclosure)	246.47	1.30292
Bulkhead 1 (rear)	164.98	0.1503
Bulkhead (front)	339.27	0.1503
Inlet Airflow Duct (including flow straightener)	494.16	0.116
Outlet Airflow Duct (including fan)	47.37	0.01575
Electronics (PCB, Arduino Nano)	331.27	0.136
HMP60 Sensor	504.16	0.028
Laser Electronics	180	~0.3
Payload Total (center of gravity, mass)	253	2.25627
Aircraft Total (center of gravity, mass)	-89.0	9.43127

Fig. 19 CDR center of gravity.

In Figure 19, the CG budget from CDR is shown. The CG must be between –88.9 and –114.3 mm for the aircraft total. In CDR, the main components were first placed where it would be most beneficial for supplying airflow the proper air, then laser electronics were placed to put the CG to the extreme. This CG budget showed that the payload was still highly adjustable and reasonably placed. In the final assembly, the electronics were not placed in their extreme positions. Some components, such as the bulkheads, had their weights reduced. Fasteners (screws, washers, nuts, and heat set inserts) were not considered in the CDR CG budget.

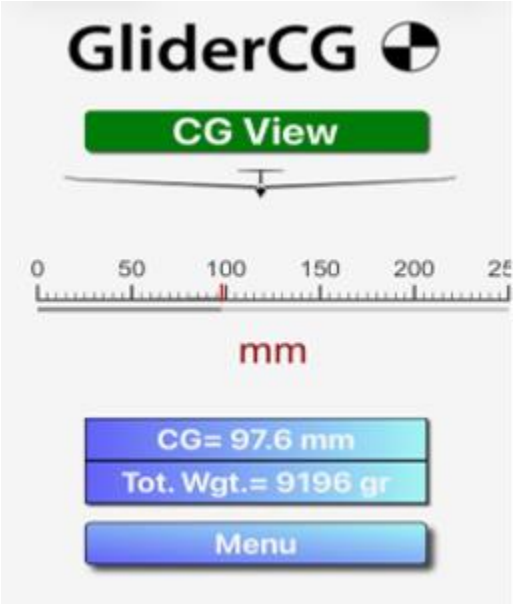


Fig. 20 Day before flight center of gravity.

In Figure 20, the center of gravity of the assembly on the drone from the day before the 4/21 flight is shown. There is an error in the CG and weight scale. The final CG from the day of the flight is 101 +/- 12 mm. These two readings are within the error range given by the +/- 12 mm. The uncertainty in the actual scale means it shows the drone to be front heavy only part of the time.

Multiple times weighing the full assembly and resetting the scale must be done, especially if the CG is shifted closer to either edge of the CG limit. The CG range is 25.4 mm but the error range is 24 mm, so extra care is needed if any other changes are made to the payload.

3.7 Airflow Configuration

3.7.1 Airflow Overview

The airflow subsystem is responsible for delivering a controlled, uniform flow of air through the laser absorption cavity to enable accurate $\delta^{18}O$ measurements. The system draws freestream air through a dual inlet assembly, conditions it through a diffuser, then honeycomb flow straighteners, turning vanes, passes it through the laser cavity, and finally expels it through an exhaust fan. The primary design objective is achieving a spatially uniform pressure and velocity field within the cavity to minimize measurement errors in the laser absorption.

To meet this requirement, each subsystem component was designed in sequence to progressively condition the flow. The diffuser expands the inlet area, reduces the velocity, and converts dynamic pressure into static pressure recovery. Downstream, a honeycomb flow straightener minimizes residual swirl and lateral velocity components inherited from the inlet. Then, turning vanes guide the flow through a bend with minimal pressure loss to the laser cavity. Once in the cavity, the air passes through the laser cavity and then gets expelled outward with the assistance of the fan. Fan speed is actively controlled via a feed-forward control law as a function of freestream dynamic pressure, ensuring the minimum required volumetric flow rate is maintained across varying flight conditions and freestream air velocities.

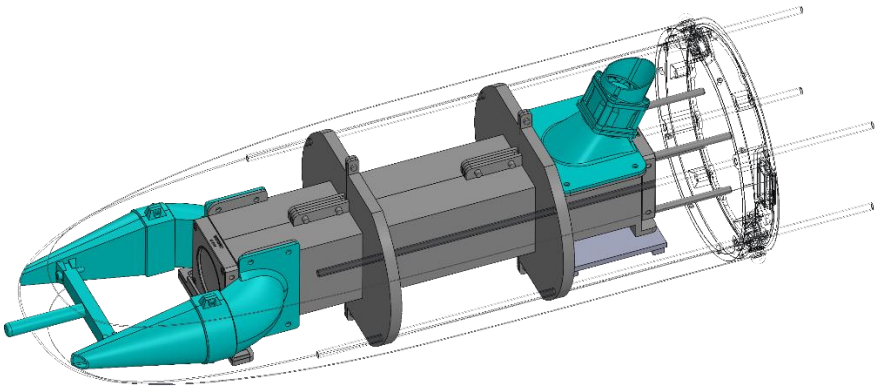


Fig. 21 Airflow subsystem isometric view.

3.7.2 Inlet

The inlet assembly serves as the entry point for the freestream air and was designed to maximize pressure recovery while minimizing flow distortion entering the laser cavity. The assembly consists of two symmetric freestream inlets mounted on either side of the instrument,

each feeding into a diffuser section that transitions the circular inlet geometry into the rectangular cavity cross section outlet.

The diffuser expands from an inlet area of 209.19 mm^2 to a cavity outlet area of 2010.26 mm^2, yielding an area ratio of 9.61. By the continuity equation for incompressible flow ($\rho = const \rightarrow V_1 A_1 = V_2 A_2$), the expansion reduces the flow velocity by approximately 90% at the cavity outlet relative to the freestream inlet velocity. This velocity reduction corresponds directly to static pressure recovery (Bernoulli's principle), converting most of the incoming dynamic pressure into static pressure available to drive flow through the cavity. The diffuser geometry was designed with a gradual expansion angle to minimize flow separation.

The inlet also incorporates a dedicated mounting interface for the HMP60 sensor from a threaded mounting point, allowing simultaneous freestream condition measurements.

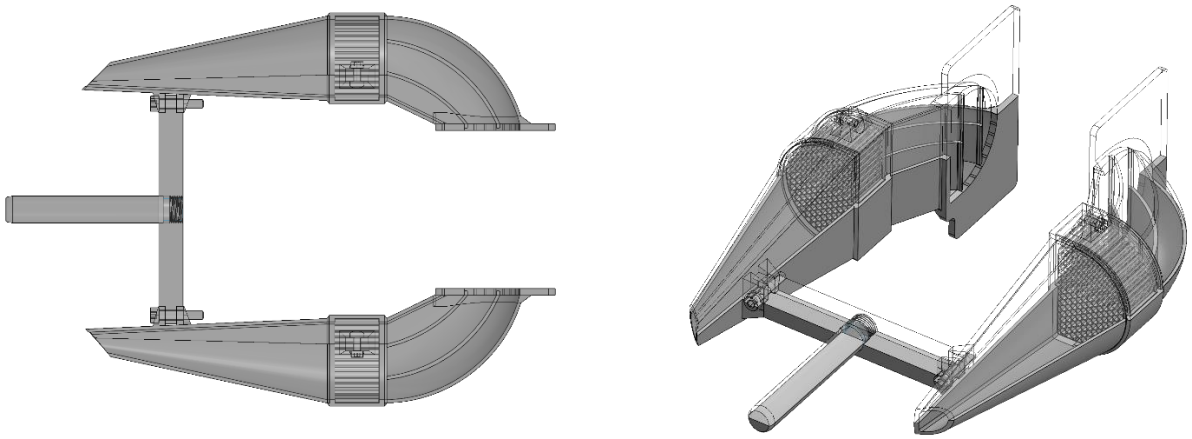


Fig. 22 Inet top and isometric view.

3.7.3 Honeycomb Flow Straightener

The honeycomb flow straightener serves to condition the flow prior to entering the turning vanes and laser cavity. It employs hexagonal honeycomb cell geometry, which provides the highest structural efficiency and most uniform flow conditioning of available cell geometries. The cell diameter to length ratio of 1:7.92 was selected based on established wind tunnel design literature, which identifies the peak performance range between 1:6 and 1:8 [4]. A ratio below 1:6 provides insufficient flow straightening, while exceeding 1:8 produces diminishing returns at the cost of excessive pressure drops. At 1:7.92, the flow straightener balances flow condition effectiveness against system pressure loss. Individual cell dimensions of 2.5 mm diameter, 19.8 mm length, and 0.75 mm wall thickness were selected to achieve this ratio within the geometric and 3D printing constraints of the inlet assembly.

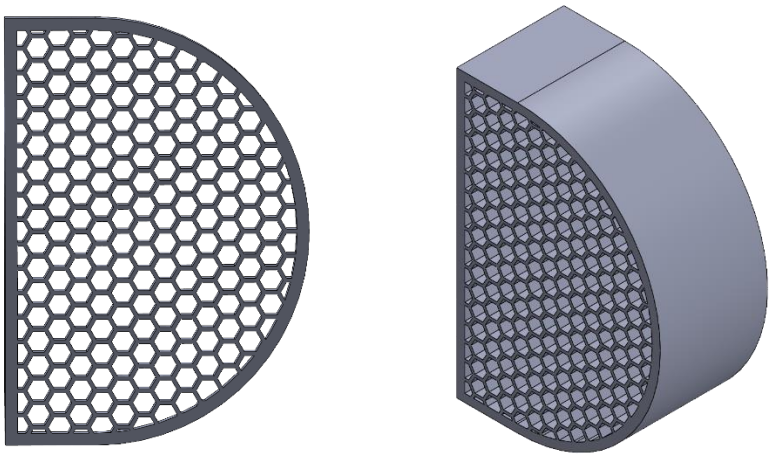


Fig. 23 Honeycomb front and isometric view.

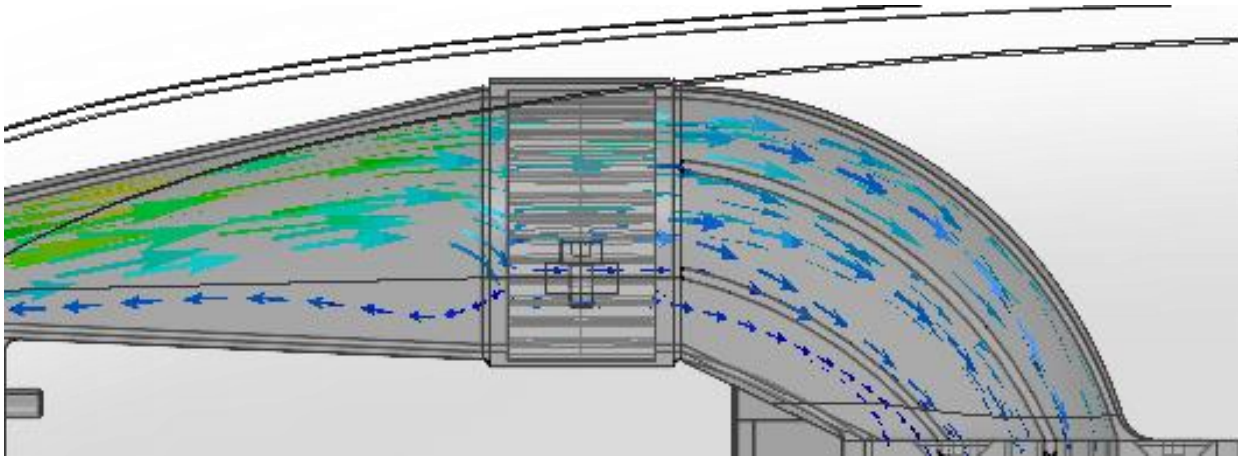


Fig. 24 Honeycomb flow conditioning.

3.7.4 Turning Vanes

The turning vanes are positioned downstream of the honeycomb flow straightener and serve to guide the flow through the direction change into the laser cavity, all while minimizing pressure losses and flow separation. Without turning vanes, bends in the flow geometry generate significant flow separation on the inner corner of the turn and produce a highly non uniform velocity profile with large turbulent eddies. The vane geometry was defined by inner and outer arc lengths of 41.55 mm and 63.91 mm, sized to span the full cross section of the inlet bend and reduce swirl entering the laser cavity. Additionally, it directly improves CFD accuracy by ensuring the inlet boundary condition more closely matches the simulation assumption of uniform flow.

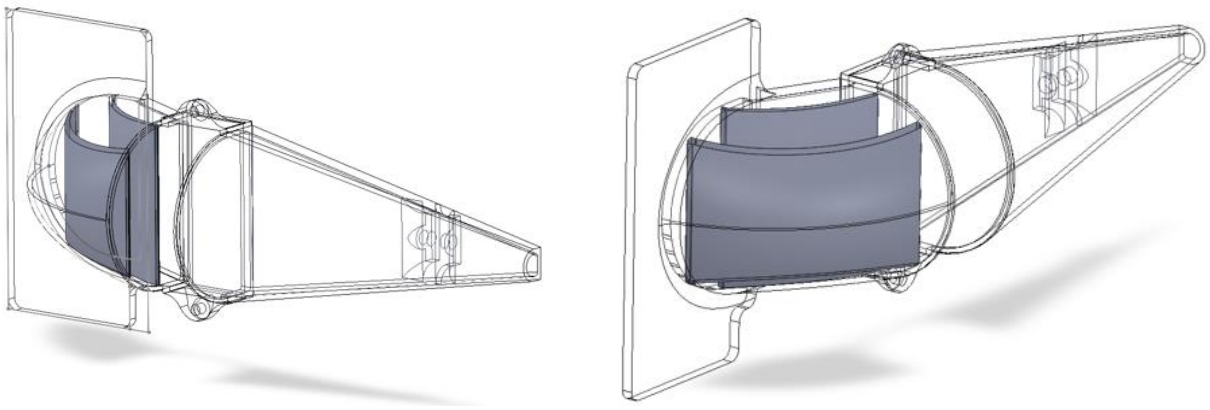


Fig. 25 Turning vanes in airflow subsystem.

3.7.5 Fan & Exhaust

The exhaust fan is the active component of the airflow subsystem, providing the pressure rise to drive flow through the cavity against the cumulative system resistance of the inlet, diffuser, flow straightener, turning vanes, and cavity. The fan was selected to be the San Ace 40 9HVA, a high performance 40 mm axial fan rated at 12 V DC with a maximum current draw of 3 A. The fan is mounted at the cavity exhaust with a custom 3D printed housing secured with M3x35 screws and M3 heat set inserts, and exhausts directly to the freestream ambient environment.

As an axial fan, it produces an annular exit flow profile concentrated in the region between the hub and rotor tip. The effective flow area is therefore defined as the annular area, $A = \pi(R^2 - r^2)$, where R is the rotor radius of 18.95 mm and r is the hub radius of 13.25 mm, yielding an effective flow area of 0.0058 m^2.

Fan speed is controlled with a pulse width modulation (PWM), with the power varying around 25% to modulate flow rate across the inlet velocity operating range of 5-25 m/s. The relationship between PWM and fan static pressure rise was characterized by the manufacturer's fan curve and interpolated across the full PWM range using MATLAB, generating a continuous family of fan curves. This interpolated fan curve database is used directly by the feed forward control law to determine the minimum PWM setting required to maintain the target volumetric flow rate at any given flight condition.

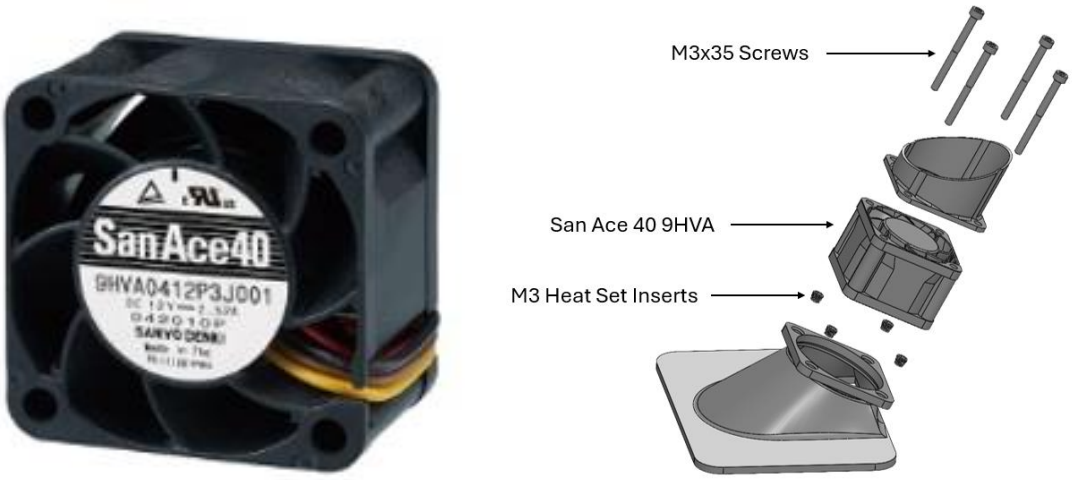


Fig. 26 San Ace 40 9HVA in airflow subsystem.

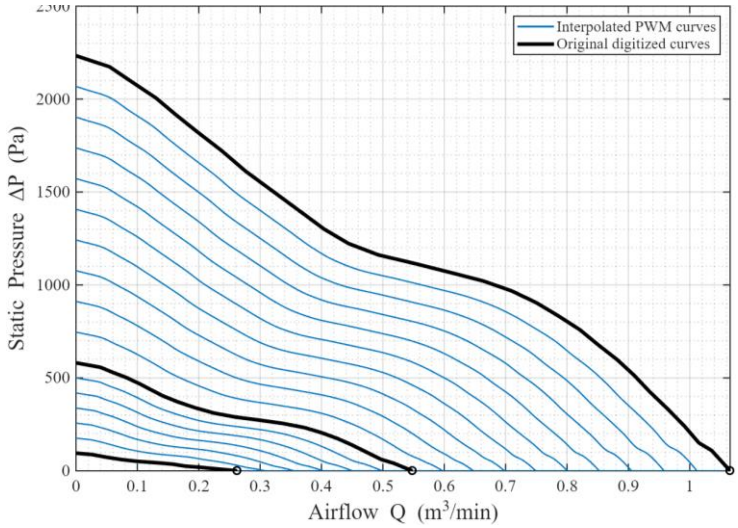


Fig. 27 Fan curve interpolation.

3.7.6 CFD Analysis

Computational fluid dynamics analysis was performed on the airflow subsystem using SolidWorks Flow Simulation to validate the inlet design and characterize the system pressure drop as a function of volumetric flow rate. The analysis was configured as an internal flow simulation with air as the working fluid, modeled as incompressible given the low flow speeds involved. The turbulence model employed was the SolidWorks modified k-ε two equation model, which automatically transitions between laminar and turbulent regimes based on local Reynolds number.

Boundary conditions were defined as swept inlet velocity ranging from 5 to 25 m/s at the freestream inlet faces, with an environmental pressure outlet of 101325 Pa at the fan exhaust. Wall conditions were set to adiabatic with a surface roughness of 25 µm, representative of the expected

3D printed surface finish of the airflow assembly. The simulation was run under steady state conditions, with convergence monitored with inlet and outlet average static pressure and mass flow rate goals. The mesh had a cell count of approximately 100,000 cells (level 5 mesh) for the first attempt, with refinements bringing the cell count up to 1,000,000 cells (level 6 mesh). They showed very close agreement in static pressure across full flow rate range, confirming that the solution was not significantly sensitive to further mesh refinement.

It is noted that the chosen boundary condition configuration imposes a fixed mass flow rate on the system, which is formally incompatible with a simultaneous fan curve matching approach. This limitation means the CFD system curve represents the inlet pressure drop as a function of imposed flow rate rather than a fully coupled fan system operating point. A pressure inlet boundary condition or external flow analysis would remove this constraint and is identified as a recommended improvement for further simulation iterations. Despite this limitation, the resulting system curve provides a valid characterization of inlet pressure drop versus flow rate, which was used directly in the fan control law development to determine required fan PWM as a function of dynamic pressure.

CFD results show the system pressure drop increases nonlinearly with the flow rate, reaching approximately 200 Pa at the maximum simulated flow rate. Velocity contour plots confirm that the diffuser, flow straightener, and turning vanes collectively produce a largely uniform flow field at the cavity inlet, supporting the core design objective of uniform pressure distribution within the laser cavity.

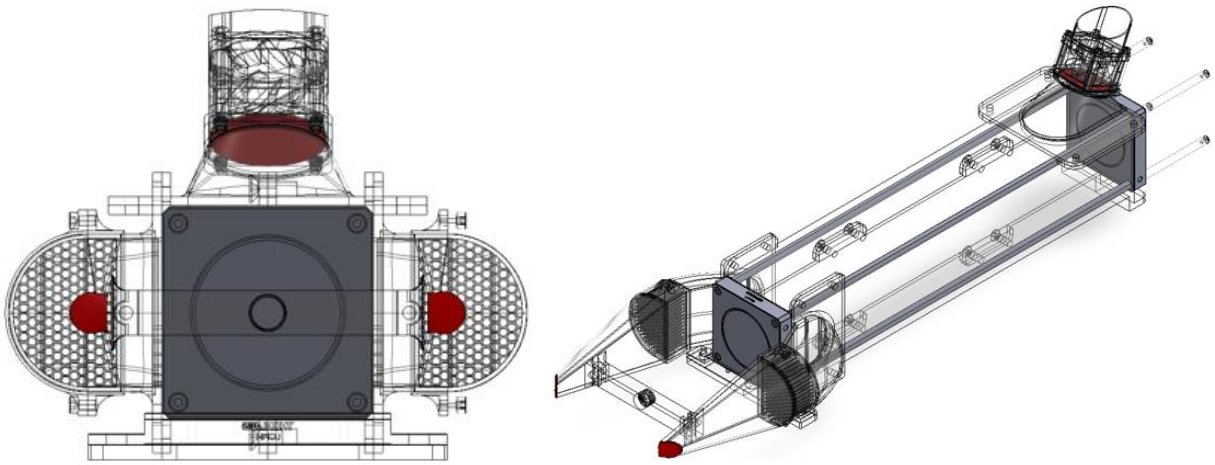


Fig. 28 Front and isometric view of CFD boundary condition locations (red).

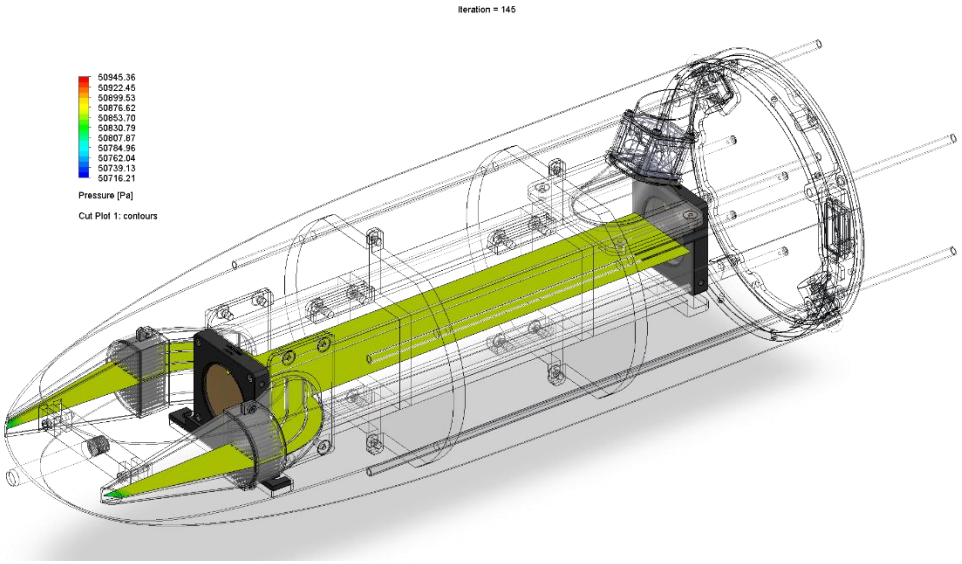


Fig. 29 Pressure plane isometric view.

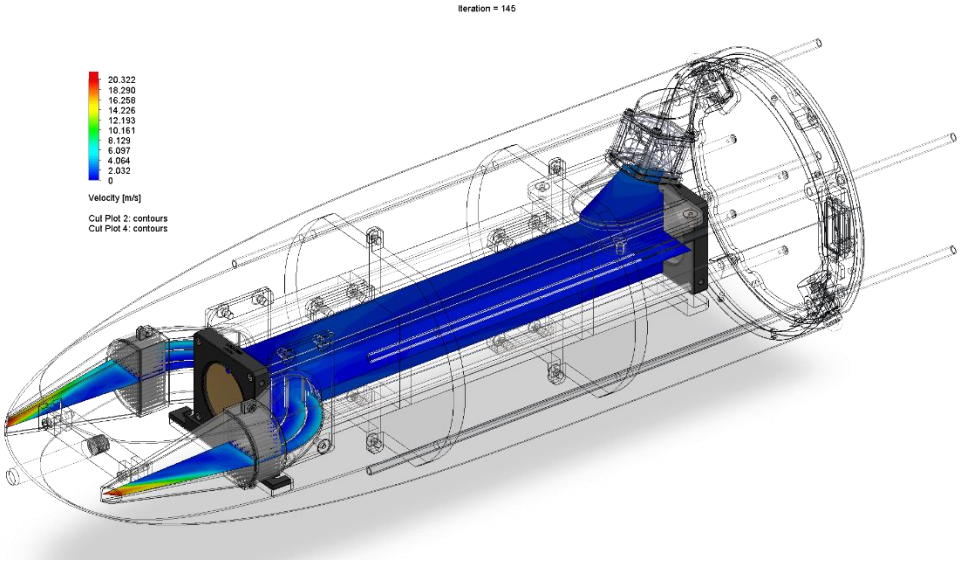


Fig. 30 Velocity plane isometric view.

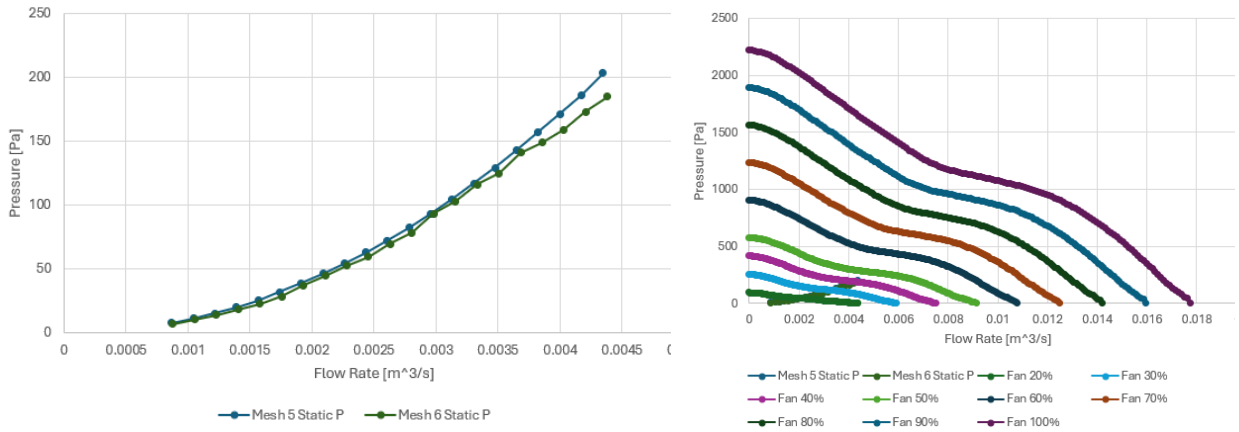


Fig. 31 Airflow system curve.

3.7.7 Testing

Experimental validation of the airflow subsystem was conducted to verify trends from the CFD derived system curve and confirm that the fan control law produces the required volumetric flow rate across the operating PWM range. Testing was performed by mounting a handheld anemometer at the exhaust fan outlet and recording the local exit velocity at discrete PWM settings ranging from PWM = 38 to PWM = 104, corresponding to approximately 14.9% to 40.8% duty cycle and measured exit velocities of 4.5 to 11.9 m/s.

Volumetric flow rate was inferred from the measured local exit velocity using the annular fan flow area, along with an empirically derived correction factor of C_f = 0.7 to account for the non uniform axial velocity profile characteristic. The volumetric flow rate was then computed as $Q = (C_f)(V)(A)$. With the flow rate established at each PWM setting, the corresponding pressure drop of the system was inferred by intersecting the measured operating point against the interpolated fan curve. Altogether, this yields an experimentally derived system curve.

The best fit experimental system curve was compared against the CFD predicted system curve, yielding a root mean squared error of 10.07 Pa with a correction factor of C_f = 0.7, indicating great agreement between the computation and experimental results and providing confidence in the validity of the CFD model and the feed forward fan control law derived from it.

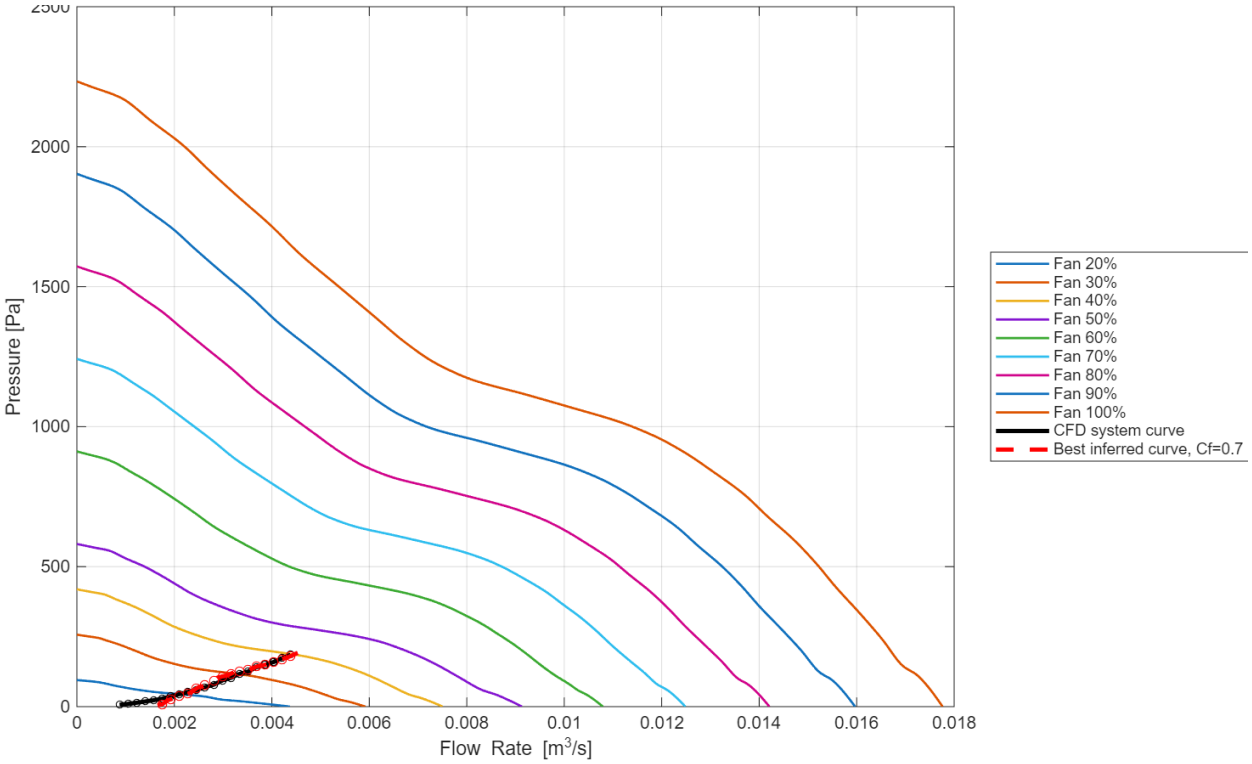


Fig. 32 Best inferred curve vs CFD system curve.

3.7.8 Fan Control Implementation

The fan control system implements a piecewise feed forward control law that determines the minimum required PWM duty cycle as a function of freestream dynamic pressure, q_∞. The control law is derived from the relationship between freestream airspeed and dynamic pressure, given by $q_\infty = \frac{1}{2}\rho V^2$, where increasing airspeed produces a ram pressure assist that partially offsets the system pressure requirement imposed on the fan. The required fan pressure rise is defined as $\Delta P_{req}(Q_{min}) = \Delta P_{sys}(Q_{min}) - \Delta P_{ram}(q_\infty)$ where $\Delta P_{req}(Q_{min})$ is the system pressure drop at the minimum required flow rate taken from the experimentally validated system curve, and ram pressure assistance is modeled as $\Delta P_{ram} = kq_\infty$ with $0 \leq k < 1$. The minimum PWM setting is then determined by finding the lowest cycle satisfying $\Delta P_{fan}(Q_{min}, PWM) \geq \Delta P_{req}$, which yields the closed form feed forward control law $PWM = f(q_\infty)$, implemented in MATLAB as a piecewise function across the full freestream airspeed operating range.

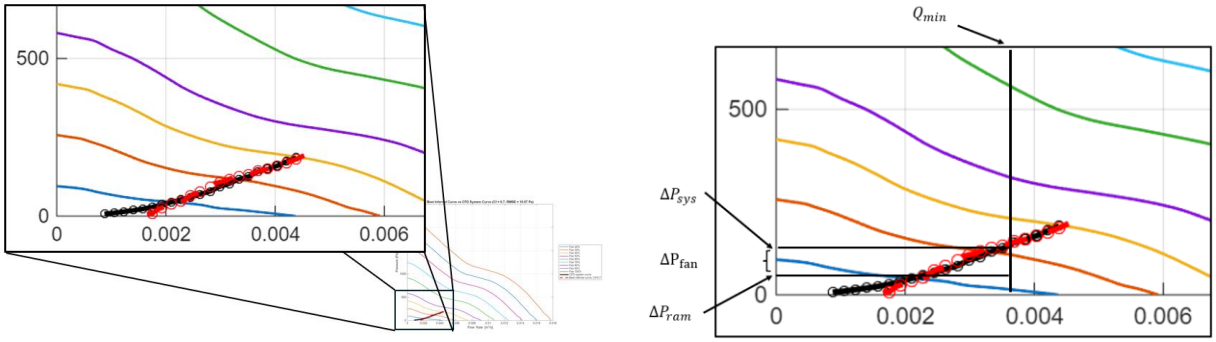


Fig. 33 PWM testing.

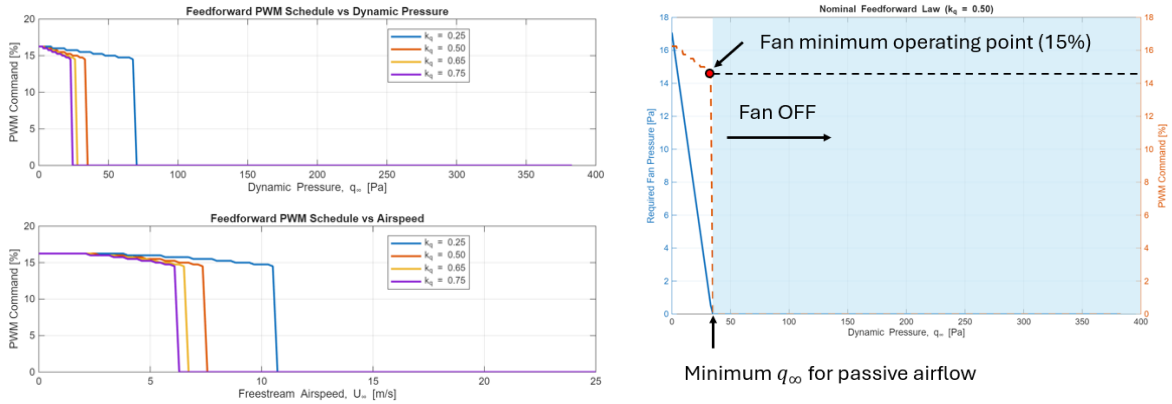


Fig. 34 Fan implementation.

The control law analysis suggests the optimal implementation would be a linear ramp down profile through the low-speed region: e.g. ramp from 20% to 15% for 0-10 m/s air speed. Any flight speeds above this *could* be driven by ram air alone, but the fan should be kept idling at its lowest setting (~15%) to avoid any restrictions imposed by a static rotor.

However, in practice it would be prudent to be conservative with fan control given the limitations of analysis and testing. These results suggest that 15-20% is the minimum necessary, however choosing a slightly higher setting of 25% or 30% would provide extra operating margin, especially in higher altitude or lower density scenarios. Additionally, the UAV communication required for the control law may be difficult. In this case, a simpler implementation would be to choose a constant fan setting, ignoring the ram air assistance.

3.7.9 Future Outlook

Current CFD and experimental testing results show good agreement on system curve characteristics, providing confidence in the validity of the flow model. For the cavity volume of v = 0.0016 m^3, approximately 200 Pa of pressure rise is required to drive sufficient flow through the system. This is readily achievable by the San Ace 40 9HVA fan operating alone, though doing so at higher PWM settings sacrifices turbulence performance in favor of meeting the minimum flow rate requirement. Future work should investigate jointly optimizing fan PWM for both flow rate and turbulence intensity, which would require additional simulation data or direct pressure transducer measurements distributed throughout the system.

A potential simplification identified for future design iterations is the removal of the ram air assist term from the control law, relying solely on the fan to drive the required flow regardless of freestream airspeed. While this would simplify both the control law and system analysis by eliminating the need to characterize the ram assistance coefficient, it would still require experimental testing and characterizations to validate fan only performance across the full operating range.

Altitude and density effects represent an open area of analysis. As air density decreases with altitude, system resistance decreases but fan effectiveness decreases by a comparable magnitude, suggesting the net effect may be near break-even. However, this has not been formally characterized and will require real flight condition testing combined with external CFD analysis to fully resolve.

Finally, pressure uniformity within the cavity has been shown to be excellent across a wide range of operating conditions. The flow sits firmly in the low turbulent regime with $R_e \approx 3000 - 5000$, which is favorable for measurement consistency. Future work should focus on experimentally confirming this uniformity within cavity instrumentation to fully validate the CFD predictions.

Chapter 4: Integration and Assembly

4.1 Fabrication Summary

Prior to finalizing the design for manufacturing, preliminary testing was conducted to validate material and structural parameters. This included UTM based infill and thickness testing to determine optimal print configurations for strength and weight considerations. Detailed procedures and results of these tests are presented in Chapter 5. Additionally, tolerance testing screws and mating holes were performed to ensure proper fit and assembly before committing full scale prints.

The majority of the components were fabricated in-house using PLA filament via fused deposition modeling (FDM). Fabrication began with the bulkheads, which were initially attempted on the INSTAAR Lab Neptune 4 Max printers. However, due to technical limitations encountered during printing, these parts were successfully produced using the ITLL Prusa MK4 printers.

For the enclosure, one unit was successfully printed on the Neptune 4 Max, while the second unit was manufactured using an Idea Forge printer to maintain production continuity. Components related to airflow management, including the flow straightener and associated inlet geometries, required higher precision due to their small, narrow, and complex features. These parts were therefore fabricated externally by a member of the team using a high precision personal 3D printer.

All 3D printed parts are designed to have as few supports as possible while maintaining a smooth surface facing the airflow. The supports were primarily tree-type supports with a top contact Z distance of 0.2 mm. When printing with the INSTAAR Neptune 4 Max Printer, we must be careful with the flow speed. The speed should be between 80-100 mm/s to avoid vibration and slipping of the axis alignment. The important parts that require external changes in slicer settings are the airflow components, particularly the flow straightener. This part needs to be printed on a 0.2 Nozzle with a reduced layer height to avoid ribbing.

Most fasteners, adhesives, and PLA filaments were procured through standard suppliers (e.g., Amazon), while select hardware components and foam padding were sourced from McGuckin Hardware to meet immediate assembly requirements.

Assembly operations were primarily conducted within the INSTAAR laboratory space. Modifications to the carbon fiber nose cone were carried out using a handheld Dremel tool in the Aerospace Wood Shop. To ensure dimensional accuracy and repeatability during cutting, a custom 3D printed jig was designed and utilized for machining the inlet openings.

Overall, the fabrication process combined iterative testing, multi-location manufacturing, and in-house assembly to achieve the final payload configuration.

4.2 Mechanical Assembly Process

This section outlines the step-by-step procedure for assembling the laser payload system. Refer to the figures throughout for visual guidance.

4.2.1 Laser and Mirror Mount Assembly

Begin by attaching the laser mount to the mirror mount plates. The steps are as follows: identify the front side of the laser (the side without the extended rods), then align the mirror mount plates with the laser mount on this front side and finally secure them using the provided screws with the mounts. Ensure proper alignment before tightening to avoid misalignment of the optical path.

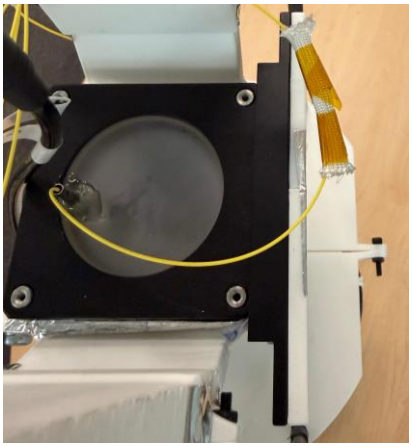


Fig. 35 Laser/mount assembly.

4.2.2 Enclosure Installation

Position the enclosure over the laser assembly. Ensure that the fan cut-out on the enclosure is oriented towards the rear side (where the extended rods are located), the optical fiber remains unobstructed and is not sharply bent, fasten the enclosure using M5 screws, and place washers before tightening the nuts to distribute load properly.

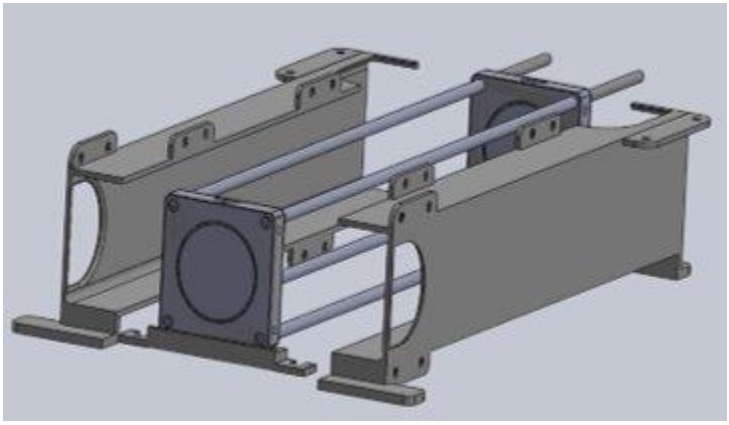


Fig. 36 CAD view of cavity.

4.2.3 Inlet Assembly

4.2.3.1 Flow Straightener Installation

Insert the flow straightener into the inlet section as shown in Figure 37, cover it with the lid structure, and secure using M3 screws with locking washers for vibration resistance. Repeat the process for both left and right inlet sides

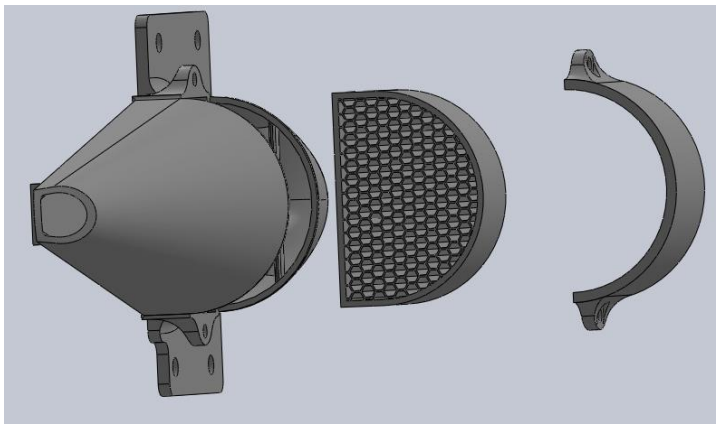


Fig. 37 CAD view of flow straightener installation.

4.2.3.2 Inlet Attachment

Attach the completed inlet assemblies to both sides of enclosure using M4 screws.

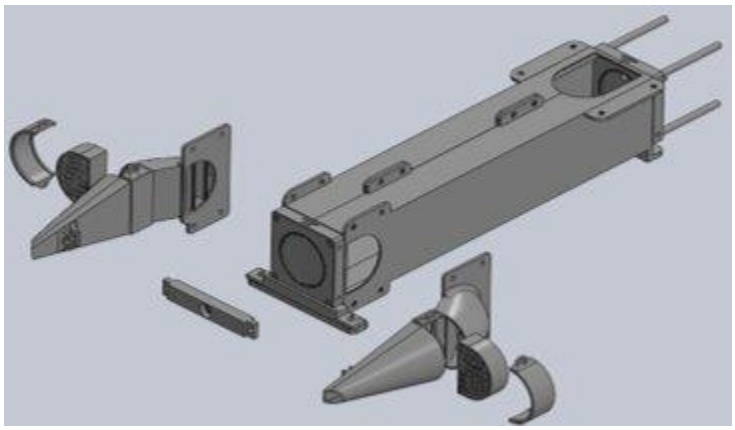


Fig. 38 CAD view of inlet attachment to cavity.

4.2.3.3 HMP60 Mount Installation

Install the HMP60 mount between the two-inlet section using M3 screws.



Fig. 39 Front view of HMP60 mount.

4.2.4 Exhaust Assembly

Attach the exhaust fan mount using M4 screws. On the left side from the inlet, either screw it from the bottom side or use shorter M4 screws so it doesn't come in the way of laser electronics. See Figure 41 for reference. This mount includes a cut-out matching the fan dimensions. Install heat-set inserts into the mount before screwing (if not pre-installed). Mount the fan onto the insert-equipped mount. Attach the airflow duct/extension (used to guide exhaust air out of the cavity). Secure the entire assembly using the longest M3x35 screws.

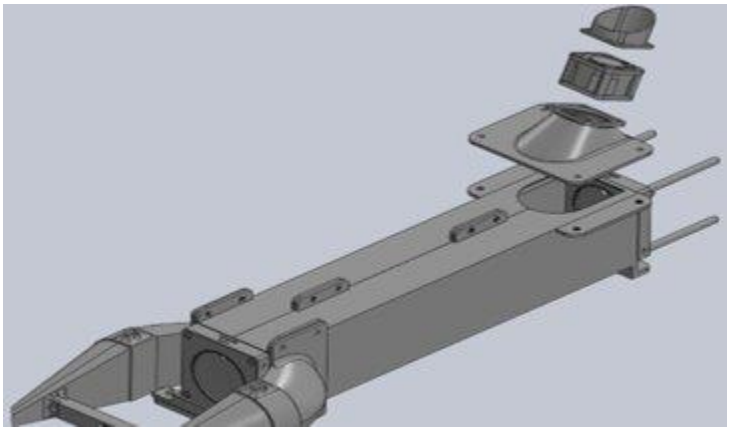


Fig. 40 CAD view of exhaust attachment to cavity.

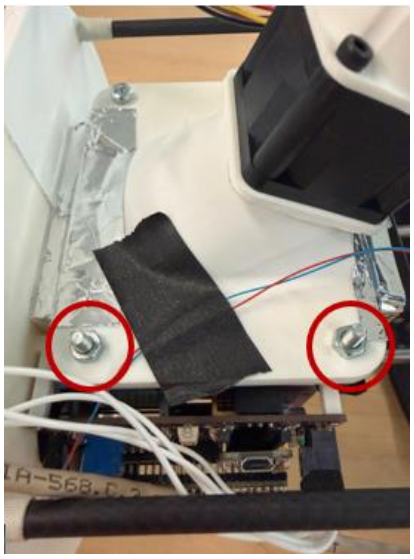


Fig. 41 CAD view of exhaust attachment to cavity.

4.2.5 Bulkhead Installation

There are four bulkheads in total. The front bulkheads have a smaller width (to fit into the nose cone region), and the rear bulkheads have a larger width.

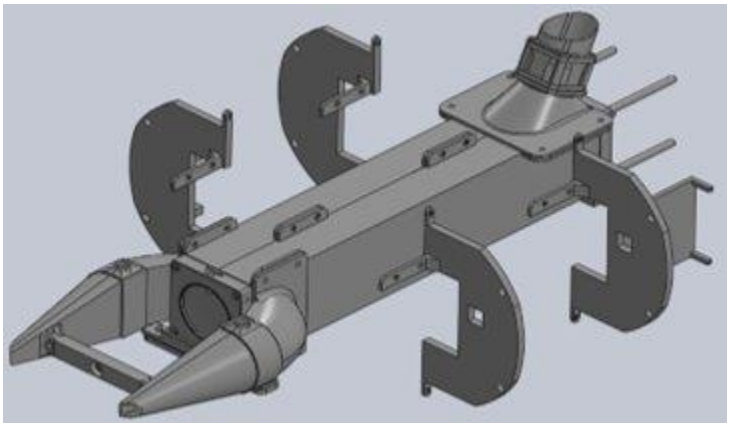


Fig. 42 CAD view of front/rear bulkhead attachment to cavity.

4.2.5.1 Front Bulkheads

Front bulkheads face the system from the inlet side. The right bulkhead has the PCB mount with soldered M3 heat-set inserts. Align both with enclosure mounting holes. Make sure PCB mount is facing outward as seen in Figure 43. Make sure to secure the M5 screws for the main attachment with the enclosure, and the M3 screws for the top and bottom connections of both bulkheads.

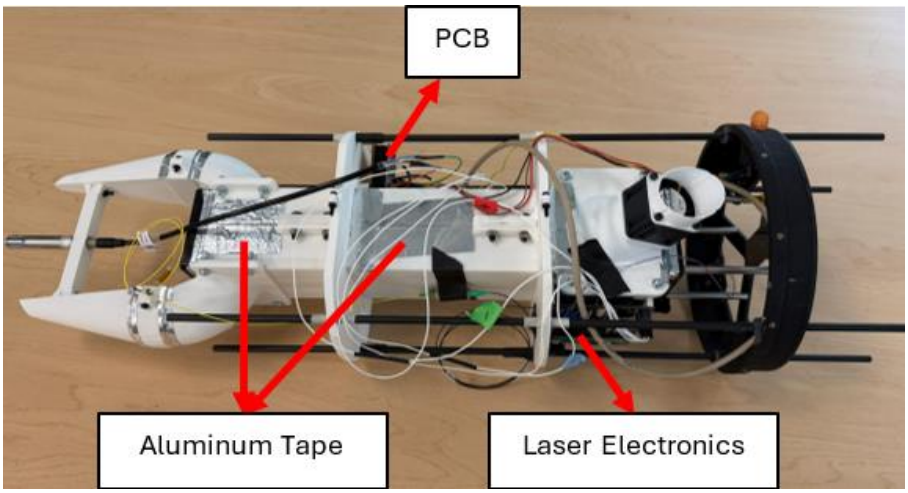


Fig. 43 Front bulkhead installation guide.

4.2.5.2 Rear Bulkheads

The left bulkhead has the laser electronics mounted with soldered 1.4 mm heat-set inserts. Install using the same fastening method as front bulkheads. Make sure laser electronics mount is facing outward as in Figure 43. Ensure all bulkheads are flushed and aligned to avoid structural stress.

4.2.6 Sealing and Vibration Control

To minimize cavity air leakage and vibration, apply aluminum tape along the enclosure seams and inlet and exhaust interfaces. Cut and install stiff foam padding on the opposite side of the laser mount, as seen in Figure 44, to reduce vibration and prevent torsional misalignment of the laser.

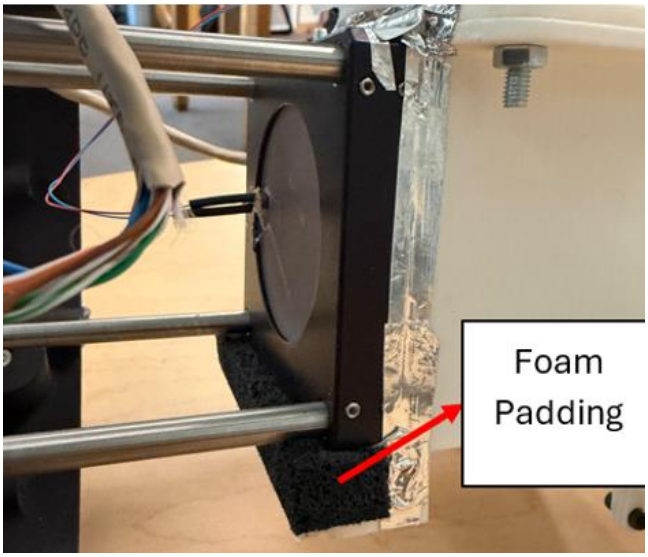


Fig. 44 Sealing and vibration control guide.

4.2.7 Final Inspection

As apart of the final inspection, verify all screws are tightened properly and confirm there are no loose components, proper airflow path, and optical fiber integrity.

For the assembly, always prioritize alignment over tightening because misalignment early will propagate errors. Additionally, handle optical fiber with extreme care, use locking washers in vibration prone regions, and avoid over tightening into plastic or printed components.

4.3 Electrical Integration

Proceed carefully to avoid damaging sensitive components.

4.3.1 Sensor Installation

When mounting the HMP60 sensor in its designated inlet mount, be aware that the 3D printed mount has threads matching the HMP60, but due to print tolerances, it may be very tight so align carefully and avoid forcing it.

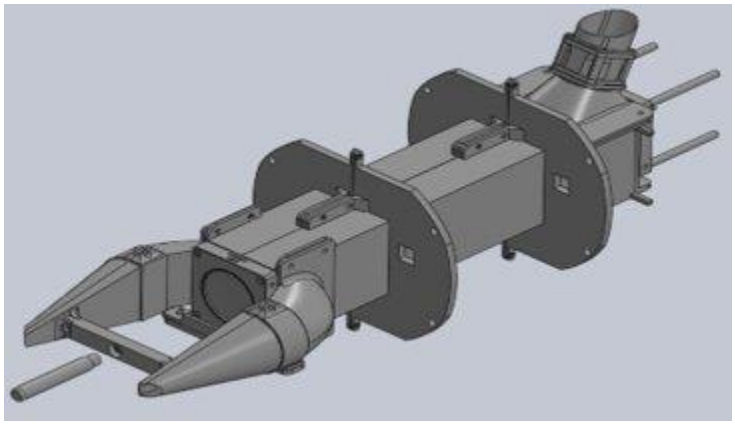


Fig. 45 CAD view of HMP60 sensor installation.

4.3.2 Laser Electronics Installation

To install the laser electronics, carefully route the optical fiber through the bulkhead opening, ensure that the SD card slot faces outward for accessibility. And secure using 1.4 mm screws into the heat-set inserts. It is critical to avoid sharp bends in optical fiber, to maintain a gentle loop (circular routing preferred), and route and secure the fiber along the enclosure wall using tape (if necessary).

4.3.3 PCB Assembly & Installation

For assembling:

- Soder to the side with the copper traces ensuring that the headers can be connected to the other side of the board (the one without any clearly visible copper traces).

- Soder any pin that has a copper connection to another point on the board. Also be sure to solder any holes that look like they are connected to the overall copper layer because these are the GND pins. Any other pin that looks like it's not actually touching any other part of copper is just for stability if you choose to solder them.
- Solder the female-male header connectors to every hole on the PCB except the larger diameter holes. Solder the 18 AWG stranded wires to the larger diameter holes and allow for enough length to connect to the buck converter (which should have electrical tape on the bottom of it to ensure that no shorting occurs).
- For the Arduino Nano's headers, solder headers such that every required pin has a header that it can plug into. In other words, if you only have 6, 8, & 10 pin headers, then it is alright to take a 6 and an 8 for each side of the board since there will be a combination such that every necessary pin is still connected to the board (since there are some middle and end pins that aren't used). If you are unsure of which pins are needed, check the bottom of the PCB or the PCB board file to figure out which pins are needed. For both sides of the Arduino Nano, using v3 of the PCB, you should be able to use an 8-pin header on the side closest to the HMP60's connector. Then leave 1 space and use a 6-pin header in the remaining spots.

For mounting the components:

- Ensure the Arduino Nano has its connector facing away from the HMP60's header slots.
- Ensure the digital logic level shifter is oriented such that the high voltage side is closest to the Arduino Nano and make sure that of the middle pair of pins on each side of the board, the GND pins are the ones closest to the edge of the PCB.
- Ensure the LM7810 has its VIN pin on the left and VOUT pin on the right. This should also result in the screw plate is facing away from the digital logic level shifter.
- Ensure the buck converter has its VIN connected to the slots located closest to the DB9 headers and that the VIN- & VOUT- are connected to the GND holes.
- For the rest of the headers (assuming you are looking at the PCB with the nano on the right and with its connector facing down), from left to right, the pins should be VIN, GND, data 1, data 2 (the fan doesn't need a data 2 connection but still has a hole for it).

For installing:

- Mount the PCB on the structure such that the Arduino Nano's connector is facing outwards (away from the main body where the laser is stored) and such that the components are facing away from the wall that the PCB is screwed into. It is critical to void sharp bends in wires, avoid loose connections in wires, and clip long header pins.

4.3.4 Final Electrical Connections

The connections are detailed in the Arduino Nano's code as comments at the top of the script. They will also be listed here:

- Notation: [Component 1] ↔ [Component 2]
    - Notation: [Component 1's connection] = [An equivalent naming convention for the connection], [Component 2's connection]
    - Notes:
        - PCB's connection is described for the corresponding header with holes numbered from left to right
        - PCB locations described assuming an orientation such that the DB9's 6-pin header is on the top left, and that the Arduino Nano is on the bottom right with its connector on the bottom side
- DB9 ↔ PCB's DB9 Header (top left 6-pin header)
        - Note: Color labels are assuming that the DB9 connector is the one used at the end of the semester
    - Pin 1 (Solid Blue) = VOUT, Pin 1 = VIN
    - Pin 2 (Striped Blue) = GND, Pin 2 = GND
    - Pin 3 (Solid Orange) UART TX, Pin 3 = Arduino Nano's built-in RX (D3 on PCB v3 but connected via soldered wire on bottom)
    - Pin 4 (Striped Orange) UART RX, Pin 4 = Arduino Nano's built-in TX (D2 on PCB v3 but connected via soldered wire on bottom)
    - [Not connected], Pin 5
    - Pin 9 (Solid Brown) UART GND, Pin 6 = GND
- Buck Converter ↔ PCB Buck Converter Holes (left side of PCB)
        - Note: The VIN side on the PCB is the one closest to the DB9's connector
        - Note: The buck converter has labels printed to the bottom of it
    - VIN+, (Top Left Top Hole) Connected to PCB's DB9 header Pin 1 via copper trace
    - VIN-, (Top Left Bottom Hole) Connected to GND via copper trace
    - VOUT-, (Bottom Left Top Hole) Connected to GND via copper trace
    - VOUT+, (Bottom Left Bottom Hole) Connected to both Pin 1s for the PCB's Fan header and PCB's Laser header via copper traces
- HMP60 ↔ PCB's HMP60 header connector (top right side above Arduino Nano)
    - VIN (Brown), Pin 1 = VIN = PCB's DB9 header pin 1
    - GND (Blue), Pin 2 = GND
    - Channel 1 (White) = Temperature, Pin 3 = Arduino Nano's A5 pin via copper trace
    - Channel 2 (Black) = Relative humidity, Pin 4 = Arduino Nano's A4 pin via copper trace
- Fan ↔ PCB's Fan header connector (in the middle below the DB9 header and above the laser header)
    - VIN (Red), Pin 1 = VIN = Buck converter's VOUT+
    - GND (Black), Pin 2 = GND
    - Brown (Controller), Pin 3 = Arduino Nano's D5 pin set to PWM mode

    - [Not used but there's a hole for it if you end up using it or just want a spot to put the yellow wire] Yellow (Sensor), Pin 4
- Laser ↔ PCB's Laser header connector (in the middle below the fan header and above the digital logic level shifter)
        - Note: Laser locations are with respect to the laser PCB provided by Dr. Darin Toohey for the project. It specifically is referring to the connector that the payload's PCB is connected to. On the laser PCB, this connector is on the bottom right side of its own board.
        - Note: The laser colors are referring to the colored tape that is on the laser's connector wires.
    - VIN (top right) (Red), Pin 1 = VIN = Buck converter's VOUT+
    - GND (top left) (Green), Pin 2 = GND
    - TX/RX (bottom two pins) (one is blue, one has no colored tape), Pins 3 & 4 = RX & TX = Arduino Nano's D8 & D9 pins respectively

4.4 Payload Integration into Black Swift S3

When installing the nose cone support rod structure through bulkhead holes, fix in place using friction tape to prevent movement. Additionally, carefully position the nose cone, ensuring the alignment of the inlet opening, exhaust outlet, and HMP60 sensor. Then seal any gaps around cut outs using aluminum tape and secure the nose cone using screws attached to the rod structure.

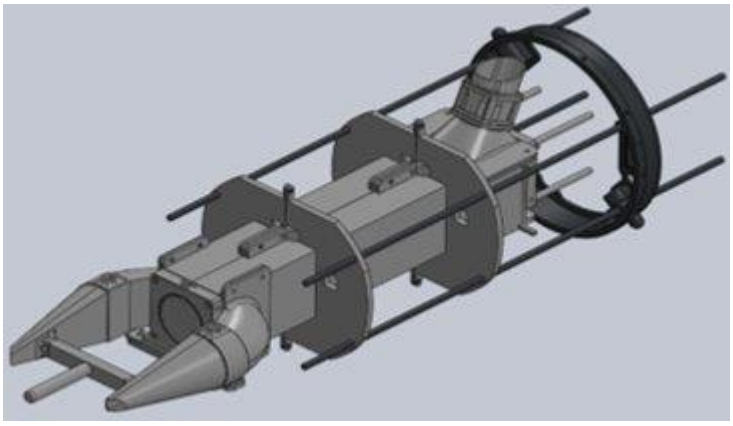


Fig. 46 CAD view of payload rod structure installation.

4.5 Integration Challenges and Resolutions

The integration phase presented several challenges spanning fabrication, assembly, and system-level constraints, requiring iterative problem solving throughout the build process.

One of the primary challenges was the fabrication of large components, particularly the enclosure, due to its extended dimensions. Printer limitations and reliability issues led to multiple print attempts across different machines. Similarly, the bulkheads required several reprints as

tolerance mismatches, and evolving design requirements became apparent during assembly. These iterations were necessary to ensure proper fit and alignment as the system matured.

During electronics integration, alignment between PCB mounting holes and their corresponding mounts proved to be non-trivial. Minor deviations from 3D printing tolerances required repeated adjustments and redesigns to achieve a secure and accurate fit. Additionally, managing fastener tolerances across different printed components introduced further complexity, often requiring careful handling and incremental fitting during assembly.

Working within the laser cavity posed another constraint, as it is a delicate subsystem sensitive to misalignment. This required careful planning of the assembly sequence, particularly when routing wiring and optical fiber, ensuring that all connections could be made without inducing stress or obstructing access. The positioning of components had to be continuously adjusted to balance accessibility, structural integrity, and functional routing of wires.

The modification of the carbon fiber nose cone also required significant attention. Due to limited tolerance for error, cutting operations had to be performed with precision, using appropriate tools and controlled methods to avoid material damage. This step demanded careful planning and execution.

Additional constraints included meeting project deadlines for ground and flight testing, while simultaneously maintaining weight limits and CG requirements. These constraints influenced both design and fabrication decisions, often requiring trade-offs between structural robustness and mass efficiency.

Fabrication of the flow straightener, with its fine honeycomb geometry, presented another challenge. Its small and complex features required the use of fine nozzle settings and precise printing conditions to achieve acceptable quality.

Finally, maintaining laser alignment during assembly and operation required special consideration. Measures were taken to minimize vibration and prevent torsional misalignment, including strategic component placement and structural support. Similarly, care was taken to ensure that optical fiber routing allowed sufficient bend radius, preventing performance degradation or damage.

All these challenges were addressed through iterative design refinement, multi-step validation, and careful assembly planning, resulting in a fully integrated and functional payload system.

Chapter 5: Verification and Testing

5.1 Electrical Testing Results

A series of tests were conducted to validate the performance and integration of the electrical subsystem. These tests were categorized into:

- **Component-level testing** (individual validation)
- **System-level testing** (integrated operation)

5.1.1 Sensor Characterization (HMP60)

The Vaisala HMP60 sensor was evaluated under controlled low-temperature conditions using the INSTAAR -20°C freezer. Testing included both passive (ambient) cooling and forced convection using a low power fan to ensure uniform thermal exposure. Measured analog output voltages were recorded across multiple temperature and humidity conditions. These data points were used to derive calibration relationships, converting sensor voltage output into physical values (temperature and relative humidity). The key observations were:

- Sensor output range: **0–5 V**
- Arduino Nano ADC resolution: **0–1023 (10-bit ADC)**
- Linear mapping established between voltage and physical measurements

This enabled implementation of conversion equations based on the minimum and maximum sensor output values, along with the corresponding ADC limits. These results confirmed correct sensor operation and reliable data acquisition within the expected voltage range.

5.1.2 Fan Control Characterization

The fan was tested independently by supplying direct power and a PWM control signal. The PWM duty cycle was varied to determine operational limits:

- **Minimum operating PWM value:** 38
- **Maximum PWM value:** 255

Below the minimum threshold, the fan failed to initiate and sustain rotation. Above this value, stable operation was maintained across the full range. These results were used to define the valid control envelope for fan operation within the control software.

5.1.3 Signal Interface Validation (Laser Integration)

Laser subsystem testing marked the beginning of integrated system validation. Testing was conducted in collaboration with Dr. Darin Toohey to verify signal compatibility and correct interfacing between components. During testing, a voltage level mismatch was identified between control signals, leading to unreliable communication. This resulted in the incorporation of a logic

level shifter (see Section 3.1.2). This directly informed a critical design update, ensuring robust signal integrity across subsystems.

5.1.4 Power System Validation (Integrated Operation)

A prolonged system level test was conducted with all electrical components operating simultaneously (excluding drone integration). The objective was to evaluate the power stability, thermal performance, and long duration reliability.

Testing revealed inefficiencies in the initial power regulation approach using linear regulators. Excessive heat generation and instability under load led to the replacement of the LM317-based design with a buck converter (see Section 3.1.2). This modification significantly improved efficiency and enabled sustained system operation.

5.1.5 Full System Integration with Drone

Final validation was performed through integration with a drone platform at Black Swift Technologies, using the S2 drone system. The electrical subsystem was connected to the onboard system, and successful operation was confirmed through stable power delivery, proper sensor data acquisition, and successful data transmission to the ground station. The ability of the ground station to record data verified end-to-end functionality of the electrical subsystem within the operational environment.

5.2 Mechanical Testing Results

Mechanical Testing was conducted to validate material selection, geometric tolerances, and structural integrity of the payload components. The testing was carried out in multiple stages, including material characterization, tolerance validation, and assembly level fit verification. Results from these tests guided the final design decisions for fabrication and integration.

5.2.1 Infill and Thickness Testing on the UTM

It was known after CDR that the mass of the payload should be reduced to create a larger margin in the mass budget. The infill and thickness tests were performed with the purpose of lowering the total mass. Both tests were performed on 3D printed dog bones in the UTM in the tension and compression modes.

The infill tests were performed first. Two sets of identical dog bones were used: the first in tension and the second in compression. Research conducted beforehand indicated infill percentages starting at 50% to be structurally sound, with lower infill percentages being undesirable for structural components. The infill percentages used for tensile loading were 15, 20, 25, 30, 40, 50, 60, and 70. The possibility of reducing the infill to around 25% was desirable but not expected. The tensile loading test revealed 50% and 60% infill could handle high loads, but 60% failure was abrupt and brittle. Infills of 30% to 50% were more linearly elastic, leading to the

conclusion that 50% was the best to balance elasticity and load bearing. Compression tests on fewer infills (due to time) revealed 50% infill to be less desirable than 40%, but since compression is less of the loading type in the payload, 50% was selected for all components.

Three dog bones of different thicknesses (6 mm, 8 mm, and 10 mm) were tested in tension. The CDR design had 10 mm thickness for the bulkheads, but this was thought to be reducible. The highest ultimate load was in the 10 mm thickness, followed by 8 mm and then 6 mm. The change in ultimate load was not linear relative to thickness, leading to a superior strength-to-weight ratio in the 6 mm thickness. Ultimately, a margin was placed on the thickness for the bulkheads used in the final assembly to make them 6.5 mm. Figure 47 shows all 3 testing types.



Fig. 47 Infill tensile, compression, and thickness tests.

5.2.2 Bulkhead Hole Tolerance Testing

The size of the bulkhead holes that interface with the drone's payload rods needed to be tested. The bulkheads had to fit onto the rods. At the time, the attachment method to keep the rods in place was glue, but due to the need to be able to disassemble the payload, this was later changed. Even with tape blocking the payload from moving, the bulkhead holes must closely match the rods. Through the test, it was realized that the same CAD model could be printed on different 3D printers, and the tolerance differences between printers would determine if the bulkhead would fit or not. Future prints requiring high precision should be done on the same printer with settings to improve precision. A final hole size was determined from repeated printing and testing.

5.2.3 Screw Hole and Fastener Testing

Figure 48 shows one of the screw hole tests that were performed. Additional tests were performed on the initial prints of the bulkheads to check alignment of screw holes. Tests revealed the amount of sizing up that needed to occur when dealing with different print tolerances of 3D printers.

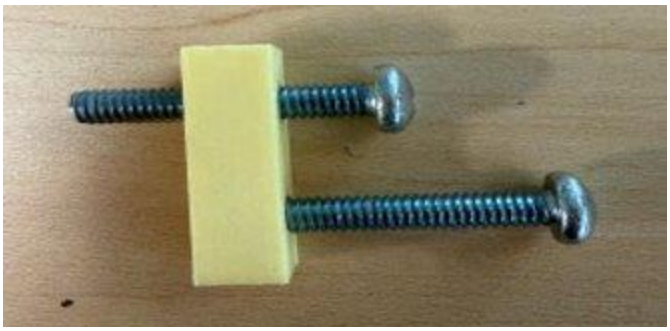


Fig. 48 Screw hole tests.

The fasteners from the electronics to the bulkheads were screws and heat set inserts. Tests were conducted on example 3D prints in order to practice heating the inserts and placing them straight and flush with the surface. The methods for holding the soldering iron, keeping the 3D print still, and fixing any mistakes were discovered. The long, thin soldering iron tip had to lay on its side if used to not burn the 3D print. The 3D print could be held by the same clamp the electrical subsystem uses for electronics. Any mistakes could be fixed by placing the soldering iron on the affected side and gently pushing down to even out the mistake.



Fig. 49 Heat set inserts tests.

5.2.4 Bulkhead-Enclosure Fit and Alignment Testing

The first test component was that the 3D printed parts would fit with the laser spectrometer and would not disrupt the instrument. The left hand side in Figure 50 shows the enclosure section fitting well onto the rods of the instrument. The same image shows the two halves of the bulkheads joining together well and fitting on the instrument properly. The second test component was that the enclosure and bulkheads would interface well without disrupting the instrument. The second image of Figure 50 shows all these components as fully assembled. Initial prints did have issues in their CAD with alignment of holes between the enclosure and bulkheads, and such issues were resolved on subsequent prints.

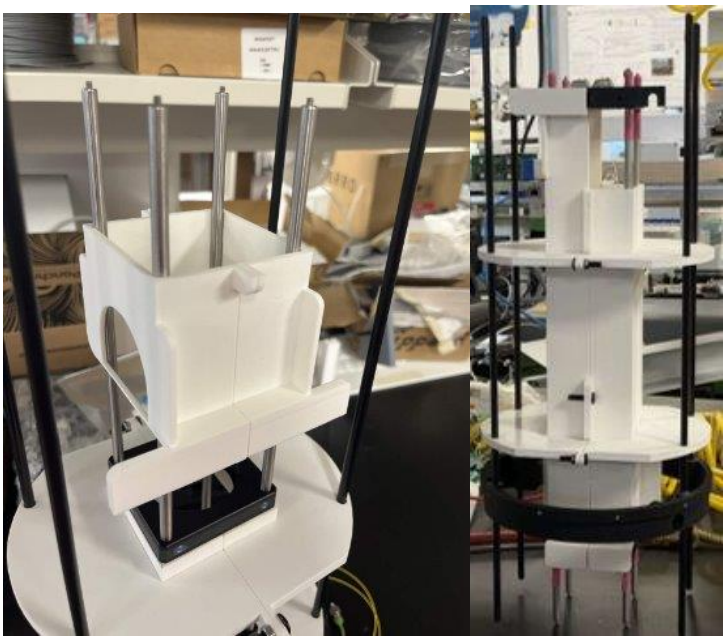


Fig. 50 Fit with laser cavity, bulkhead enclosure fit.

5.2.5 Assembly-Level Fit Testing

Once all the components were acquired or manufactured, the final assembly and tests were conducted. The assembly was done in the manner described by Chapter 4. During assembly, the main anomaly was on the interface between the enclosure and the outlet of the airflow. One of the electronics boards interferes with where the screws for the enclosure outlet interface needed to be. The solution was to flip the assembly and screw in upside down to leave room for the electronics board. All other interfaces and alignments were properly designed and printed.

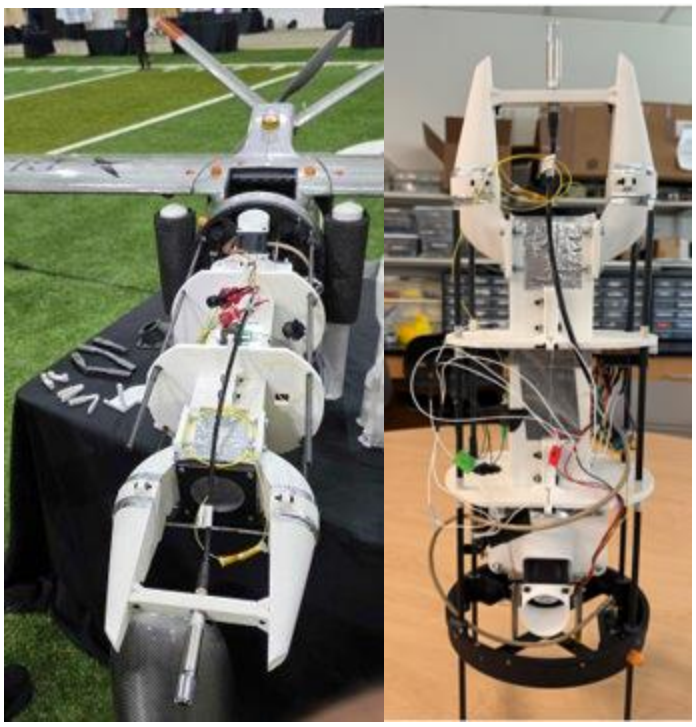


Fig. 51 Full assembly images.

Chapter 6: Budget Analysis

6.1 Cost Budget Analysis

The FLAIR project operated under a total allocated budget of $2,000 USD. Across the project lifecycle, purchases were made on eight separate occasions, spanning from an initial purchase order in January 2026 through final PCB procurement in April 2026. The total amount utilized was $945.13, leaving a remaining margin of $1,054.87.

The largest single expenditure was the Vaisala HMP60 Humidity and Temperature Probe at $403.00, reflecting its role as the primary environmental sensing component of the payload. The next most significant cost category was the laser mounting hardware, with two mounts for the laser setup totaling $200.00. The axial fan was procured for $45.00. Printed circuit board development iterated through three versions, with PCB v1, v2, and v3 costing $13.91, $19.07, and $17.22 respectively, reflecting the incremental refinement of the electrical design across the ORR phase. PLA filament was purchased on two separate occasions totaling $87.96 across both orders. Structural and assembly hardware, including screws, nuts, washers, heat set inserts, and JB Weld plastic adhesive, accounted for approximately $60 in aggregate. Optical components including 2" mirrors were procured for $15.00. Sealing and surface preparation materials including silicone sealant, foil tape, sandpaper, and weather stripping accounted for an additional $18.11. A 3D FDM Prusa laser enclosure print through ITLL was obtained for $8.03. The full itemized breakdown is provided in Table 3.

Table. 3 Cost budget breakdown

Component	Qty	Unit Cost (w/o taxes)	Total Cost
HMP60 Humidity and Temperature Probe	1	$403	$403
Axial Fan	1	$45	$45
Mounts for laser setup	2	$100	$200
General Screws	1	$8.69	$8.69
General Washers	1	$7.69	$7.69
General Nuts	1	$8.99	$8.99
Heat Set Inserts	1	$8.99	$8.99
Heat Set Inserts Screws	1	$6.99	$6.99
JB Weld Plastic Adhesive	1	$7.73	$7.73
PLA	2	$21.99	$43.98
M3 Screw Kit	1	$22	$22
M3 - M5	1	$20	$20
Silicone Sealant	1	$14	$14
Mirrors 2", 3mm Pack of 6	1	$7	$15
PLA	2	$21.99	$43.98
Sandpaper AL Oxide	3	$1.79	$5.37
Weather Stripping	1	$4.25	$4.25
Foil Tape IPG	1	$4.49	$4.49
Washer M4	15	$0.19	$2.84
Nut Hex C8	15	$0.23	$3.38
Screw (0.7x16 mm)	15	$0.34	$5.13
Weather Stripping	3	$1.39	$4.17
Screw (1.6x12mm)	2	$0.68	$1.35
Printed Circuit Board (PCB) v1	1	$13.91	$13.91
Printed Circuit Board (PCB) v2	1	$19.07	$19.07
3D FDM Prusa Laser Enclosure Print	1	$8.03	$8.03
Printed Circuit Board (PCB) v3	1	$17.22	$17.22
		Total	**$945.13**

6.2 Mass Budget Analysis

The FLAIR payload mass budget was developed with the Black Swift S3 payload capacity as the primary constraint, with the total allocated mass of 2,267.96 g. The total estimated mass of all payload components is 2,086.917 g, yielding a remaining margin of 181.043 g. It should be noted that the payload mass also falls within the standard mass allocation for the Black Swift S2 platform, confirming compatibility across both vehicle configurations.

The most mass-significant components are the laser cavity enclosures, with two units contributing 462.96 g combined, followed by the mirror rails at 416.00 g across four pieces, and

the bulkheads and component panel at 300.60 g across four units. Mirror holders account for 208.00 g across two units, and the inlet airflow ducting contributes 1.557 g. Smaller contributors include the airflow fan at 57 g, PCB frames at 75 g, and laser mounts at 80 g. The electronics, including the Arduino Nano, PCB, HMP60, and associated wiring, account for approximately 80 g in aggregate. Fastener hardware including M3 screws, nuts, and washers contributes approximately 34 g. The full component-level mass breakdown is provided in Table 4.

Table. 4 Mass budget breakdown

Component	Quantity	Unit Mass (g)	Total Mass (g)
HMP60 Humidity and Temperature Probe	1	28	28
Arduino Nano Board	1	7	7
Digital logic level shifter	1	1.25	1.25
PCB	1	30	30
PCB Frames	1	75	75
Voltage divider (LM7810)	1	1.25	1.25
Mirrors	2	68	136
Mirror holders	2	104	208
Mirror rails	4	104	416
Laser "box"	2	231.48	462.96
Mounts for laser setup (clamps)	2	40	80
Bulkheads / Component Panel	4	75.15	300.6
Airflow "ducting" (Inlet)	1	116.557	116.557
Airflow "ducting" (Exhaust)	2	7.875	15.75
Airflow fan (pump)	1	57	57
Laser electronics	1	100	100
Screws (M3)	38	0.5	19
Nuts (M3)	38	0.4	15.2
Washers (M3)	32	0.15	0.15
Buk Converter	1	17.2	17.2
		Total (g)	2086.917
		Allotted (g)	2267.96
		Margin (g)	181.043

6.3 Volume Budget Analysis

The FLAIR payload volume budget was developed against the internal volumetric capacity of the Black Swift S3 nose cone payload bay, with a total allocated volume of 1,800 cm^3. The

volume budget accounts for the block volumes of the four major payload assemblies: the laser cavity, inlet assembly, exhaust assembly, and bulkheads. The total volume utilized across these major assemblies is 636.77 cm^3, yielding a substantial remaining margin of 1,163.23 cm^3.

It should be noted that the block volume budget encompasses only the major structural and optical assemblies. Smaller components such as electronics, fasteners, and wiring harnesses are not individually captured in the block volume accounting, as they occupy interstitial space within the overall payload envelope rather than defining the primary volumetric footprint.

The laser cavity is the largest single volumetric contributor at 162.546 cm^3, followed by the bulkheads at 266.64 cm^3 across four units, and the inlet assembly at 157 cm^3. The exhaust assembly contributes 50.58 cm^3. The full volume breakdown is provided in Table 5.

Table. 5 Volume budget breakdown

Component	Quantity	Block volume (cm^3)	
Laser Cavity	1	162.546	
Inlet Assembly	1	157	
Exhaust Assembly	1	50.58	
Bulkheads	1	266.64	
		Total (cm^3)	636.766
		Allotted (cm^3)	1,800
		Margin (cm^3)	1,163

6.4 Power Budget Analysis

The FLAIR power budget was developed against an allocated total of 66 W, based on the power available from the Black Swift S3 avionics supply. The payload draws power from four primary electrical consumers: the axial fan, the HMP60 sensor, the Arduino Nano, and the voltage divider network.

At maximum expected operating conditions, total power consumption reaches 44.98 W, yielding a margin of 21.02 W. At minimum expected operating conditions, total consumption is 32.56 W, yielding a margin of 33.44 W. The axial fan is the dominant power consumer, drawing between 30 W and 35 W depending on speed setting, and represents the primary driver of both minimum and maximum power figures. The HMP60 draws between 0.66 W and 0.88 W across its operating voltage range of 8–28 V. The Arduino Nano consumes between 1.9 W and 2.5 W operating between 7–12 V. The voltage divider network has a wide operating range of 0.00144 W to 6.6 W depending on configuration, reflecting the variability in its resistive load across operating conditions. The full power budget breakdown is provided in Table 6.

Table 6. Power Budget Breakdown

	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Voltage Divider	10	5000	0.144	1.32	0.00144	6.6
HMP60	1	5	8	28	0.66	0.88
Arduino Nano	19	195	7	12	1.9	2.5
Axial Fan	120	2600	10.2	13.8	30	35
				Total (W)	32.56144	44.98
				Allotted (W)	66	66
				Margin (W)	33.43856	21.02

Chapter 7: Communications and Ground Operations

7.1 Telemetry Architecture Verification

The final version of the telemetry architecture as of the end of the semester is as follows:

- Notation: [Indices](Size in bytes) Name, Values/Notes

1. [0](1) HDR0, “U” == 0x55
2. [1](1) HDR1, “$” == 0x
3. [2](1) Type, a number from 0 through 255 (though only 0 through 5 have meanings)
4. [3](1) Action, a number from 0 through 255
5. [4-5](2) Size, max value is 128
6. [6-9](4) To address, [0xFF, 0xFF, 0xFF, 0x41]
7. [10-13](4) From address, [0x34, 0x12, 0x00, 0x00]
8. [14-46](21) Data
    a. [14-17](4) Arduino Nano loop start time, in ms
    b. [18](1) Arduino Nano packet number, a number from 0 through 255
    c. [19-20](2) HMP60 temperature, a number from 0 through 1023
    d. [21-22](2) HMP60 relative humidity, a number from 0 through 1023
    e. [23-26](4) Laser loop start time, in ms
    f. [27-30](4) Laser packet number, a number from 0 through 2^(32) == 4,294,967,296
    g. [31-34](4) Laser one second mixing ratio
    h. [35-38](4) Laser one second background
    i. [39-42](4) Laser one second peak
    j. [43-46](4) Arduino Nano loop end time, in ms
9. [47-48](2) Fletcher 16-bit Checksum

When a packet is received, several values are checked. First, the first two bytes must be "U$" and the fourth should have a value of 0. Second, only specific types have corresponding code implemented since not all packet types are needed. Third, after the total packet has been received, the 16-bit Fletcher checksum is calculated and compared to the one that was received. If any of these conditions aren't fulfilled, then the packet is thrown out.

7.2 GUI Operational Validation and Data Storage Verification

The verification for the Ground Station User Interface involved establishing connection with the payload code that is going to extract data and send to the ground in the form of data packets. To do that, we had to establish a connection between the Arduino code and the Python Ground station through a common Wi-Fi connection.

During testing, we faced various issues with showing the connection and sending the packets to the Ground station. As the timeline bottlenecked towards the ORR and the flight data set, we had to resort to a different approach, along with this connection to provide a closure to both the data analysis. That is also mentioned as an upgrade in Section 3.5.2. We resorted to an offline data-loaded post flight analysis. It works exactly like the real time telemetry plotting, with the main difference being that it takes the data from a text file after the flying procedures, pulled from the onboard SD card, and then parses through the data to plot to the required level of fidelity and accuracy. After the test flight on Tuesday, we received close to 20,000 packets during our 20 mins test flight. Once the flight payload was safely recovered, we pulled the data from the SD Card and used it to run this GUI plotting stage, to generate some exciting and interesting plots.

The future stages of this work involve proving and testing the secure connection, and live plotting during the testing and aircraft run phase, rather than just using the post flight analysis window.

Chapter 8: Risk Closure and Safety

8.1 Updated Risk Assessment

The updated risk assessment reflects the transition of the project from the design phase at CDR to fabrication, assembly, and integration leading into ORR. While the CDR risk matrix, seen in Table 9, primarily captured component-level and process-driven risks, such as manufacturing constraints, supplier delays, and testing availability, the current risk matrix, seen in Table 9, focuses on operational and system-level risks that directly impact performance during deployment.

At this stage, the dominant risks are no longer related to feasibility or resource availability, but to integration of fidelity and functional reliability. Specifically, risks such as structural integrity under vibration, optical alignment stability, airflow quality within the enclosure, and electrical

reliability are now prioritized because they have a direct effect on data accuracy and mission performance.

This shift is expected as the system moves from a conceptual and fabrication-driven phase to a hardware-realized, performance-critical phase, where even small deviations (e.g., air leakage, misalignment, CG offset) can significantly degrade system output. The updated assessment therefore emphasized risks that influence measurement accuracy, stability, and repeatability, rather than buildability.

All mitigation strategies listed in Table 9 are grounded in validated testing and assembly controls (e.g., UTM testing, dry-fit validation, airflow checks), reflecting increased confidence in risk characterization compared to the assumption-based mitigations at CDR.

Table 7. Risk Assessment

Risk	Impact	Mitigation
Laser misalignment due to vibration	Data loss/inaccurate measurements	Rigid bulkhead and enclosure mounting, clamp on one side and foam padding on the other
Fastener loosening (vibration)	Structural Instability	Use locking washers, torque checks before flight
3D printed part failure (PLA)	Structural/component failure	Optimized infill & thickness (UTM tested), safety margins applied
Air leakage in enclosure	Disturbs flow, introduces turbulence → reduces spectroscopy accuracy	Seal joints (aluminum tape), tight assembly, airflow check
Improper CG / overweight	Flight instability	Mass tracking and CG validation during assembly
Nose cone cutting error	Misalignment/poor fit	Use jig, careful cutting, dry fit
Optical fiber bending/damage	Signal degradation or failure	Maintain minimum bend radius, secure routing along enclosure

Electronics overheating	Sensor/laser malfunction	Voltage regulation ensures stable power supply to active cooling system
Lose/unsoldered wires	Signal loss/failure	Secure + verify connections

8.2 Risk Retirements Since CDR

Since the CDR, several risks identified in Table 10 have been retired or significantly reduced as the project progressed through fabrication and testing.

Primarily, manufacturing and resource-related risks, including 3D printer limitations, material availability, testing facility access, and supplier delays, have been effectively retired. These risks were mitigated through successful fabrication across multiple facilities, early procurement of materials, and completion of required testing (e.g., structural validation), thereby eliminating uncertainty associated with build feasibility and schedule dependencies.

Additionally, risks associated with component transportation and handling have been minimized due to localized fabrication and controlled assembly processes.

Some risks, such as mass growth and center of gravity deviation, as well as human assembly error, have been reduced through iterative validation (mass tracking, dry fits, use of jigs), but remain partially active due to their dependence on final system integration.

At the same time, the transition beyond CDR has introduced new integration-driven risks, captured in Table 10, which were not dominant earlier. These include risks related to enclosure sealing, vibration-induced misalignment, optical fiber handling, and electronics thermal performance. These are inherent to system-level realization and are being actively managed through assembly procedures and validation checks.

Overall, the project has successfully retired design feasibility and infrastructure-related risks, with the remaining risks concentrated at the operational and integration level, consistent with the system's progression toward flight readiness.

Table 8. Electrical Testing Risk Retirements

Component	Test Planning	Testing Parameters	Test Design	Solution to Failure
Electrical Testing				
Voltage Divider	Checking the accuracy using a multimeter	V_out is within 5% of the expected value.	Measure resistor values with a multimeter, apply	If a resistor (or other component) has a wrong value, get a new

	and power supply.		input voltage using the power supply, and verify the voltage output is within the expected range.	resistor (or other component) from the aerospace electronics shop.
PCB	Measuring electrical continuity using a multimeter.	The board is able to sustain up to 25.2V and 3A without overheating	Use a multimeter to test the continuity of the PCB.	If there is a failure in the PCB, then reprint it.
HMP60 Sensor	Confirming the temperature/RH accuracy outside.	The HMP60 is able to measure temperature to within +/- 5 degC & +/- 5% RH.	Use a computer and HMP60 with a power supply to test temperature and RH under various conditions.	If the HMP60 fails or doesn't work, then call the supplier and/or buy another one.
Fan (San Ace 9HVA0412P 3J001)	Check if the fan runs when power is provided.	Rated Input of 1.4 - 31.2 W	Use a video camera and power supply to measure the RPM of the fan.	If the fan fails or doesn't work, then call the supplier and/or buy another one.
Laser	Check if the laser provides data when power is provided.	The laser is able to transmit data and be received by another system to decode.	Measure laser output with a power supply and computer in different conditions to verify functionality.	If the laser fails or doesn't work, then call Dr. Darin Toohey to debug and figure out what the issue is in order to get it to work.
Payload Code	Ensure we can receive and send data from the NANO.	The payload code is able to communicate between each device (not including to the ground station).	Use a breadboard, jumper wires, a power source, and a computer to verify the circuit functions correctly before connecting it to the PCB.	If the payload code doesn't work, then use a computer with Arduino installed to debug it.

Comms Test	Ensure we can send and receive information between the systems.	The drone is able to communicate to the ground station.	Test data transfer between the drone and ground station using two computers, a BlackSwift S2/S3, and a Wi-Fi router.	If the drone cannot send/receive messages, contact BlackSwift Technology and debug the code to identify the issue.
	Measure the accuracy by checking missed data packets.	Verify the drone can transmit all data, or store 90% of any missed packets onboard.	Use post processing from the previous test to determine how many packets were missed.	If packets are missed, debug the code, consult BlackSwift Technologies, and store data on the payload for post-flight analysis.

Table 9. Mechanical Testing Risk Retirements

Component	Test Planning	Testing Parameters	Test Design	Solution to Failure
Mechanical Testing				
Material - PLA infill	Determine how infill percentage affects strength and stiffness	UTS (MPa), Young's modulus (GPa), elongation at break (%), full stress-strain curve, failure mode	PLA dog bone with different infill% (30, 50, 60, 75)	Increase print temperature, add more perimeters, change infill pattern, dry filament, or increase infill for weak regions
Epoxy glue Testing (PLA-PLA & PLA-	Evaluate bond strength and best surface preparation method	Peak shear strength (MPa), load-displacement curve, failure location/type(adhesive/cohesive/substrate), bondline thickness	Create lap-shear coupons and test on UTM to measure shear	Improve surface prep/primer, increase overlap or bond area, change

Carbon Fiber)			strength and observe failure type	adhesive, add mechanical backup
Tape Testing	Validate tape for sealing or temporary bonding	90° peel force (N/cm), static shear-holding time under load (hours), adhesive change after temp/humidity (% loss)	Perform 90° peel tests and shear-load tests; check adhesion under heat/humidity.	Use stronger tape, clean surfaces better, increase contact area, or replace tape with mechanical fastening.
Gluing the foam	Verify that selected adhesive does not thermally or chemically damage foam under worst-case bond volume.	Peak foam surface temperature (°C), visual damage, dimensional change, and bond strength.	Thermocouple-monitored foam-to-PLA coupons (nominal + worst-case bond volumes), cure log, then peel/tensile check (n=3).	Switch to low-exotherm or foam-safe adhesive, reduce bond volume/increase area, add barrier layer, or use mechanical retention.
3D-Printed Bulkhead (PLA)	Tensile Compression Vibration Testing Volume Measurement	400 N tensile 600 N compression 20–2000 Hz at 1.5 g Tolerance ±0.3 mm	Print 3 samples UTM load to failure Vibration sweep Dimensional check	Increase infill Switch filament Add fillets Alter CAD mode/reprint
Adhesive + PLA Material	Shear and tensile characterization	>8 MPa shear 300 N tensile 50–80% RH environment	Lap shear coupons Cure 24/48 hr Shear test in UTM	Surface prep Longer cure Stronger adhesive

Bulkhead Structural Test	Static load and modal testing	500 N static load Deformation <1 mm 20–200000 Hz vibration	Load increments of 100 N Modal frequency sweep with sensors and accelerometers	Add ribs, increase thickness

Chapter 9: Conclusions and Path Forward

9.1 ORR Outcomes

9.1.1 Airflow

Analysis and testing of the airflow subsystem strongly suggest that the as-built design meets the minimum flow rate requirement through ram air assistance and fan power. In practice, the fan was set to a constant speed to guarantee flow rates in any flight condition.

The robust design of the airflow system ensures that pressure is uniform throughout the cavity regardless of flow conditions at the inlet. The honeycombs effectively smooth out any jetting or other non-uniformities, restricting their influence to the diffuser section. Furthermore, the flow area match between the inlet and laser cavity ensures the flow is minimally perturbed after the honeycomb section. Once the flow has been conditioned, there are no further area manipulations, therefore preserving pressure uniformity.

An open question remains around the amount of turbulence within the measurement cavity, and if that has a significant impact on measurement accuracy. CFD analysis suggests that the flow will be slightly turbulent, and further testing is required to understand the ramifications imposed.

9.2 Remaining Work

9.2.1 Payload Software

There are two parts of the payload software that still have remaining work.

First, the Arduino Nano should eventually also be able to communicate data to the laser PCB and not just from it. This would allow the laser to modulate itself according to the physical data collected by the rest of the system including temperature, relative humidity, pressure, wind speed, etc. This would also allow the laser system to store this data in its SD card for post flight analysis if there is a breakdown of communications.

Second, the Arduino Nano should be able to receive information from the drone's DB9 connection. This will allow the ground station to send commands to the payload and would also enable the payload to collect relevant data from some of the drone's built-in sensors like the static and dynamic pressures.

Following these changes, the electrical system's PCB will be updated and optimized to account for any new designs and more spacing.

9.2.1 Airflow Control

In its current configuration, the airflow subsystem is completely functional but may not guarantee accurate measurements. Further integrated testing should be conducted to determine if isotope measurements are sensitive to higher turbulence and flow rates. If the sensitivity is acceptable, the system may be operated in its current state, with a constant fan speed for all flight conditions.

If the sensitivity is not acceptable, the airflow system must be tested and tuned to provide flow rates closer to the minimum necessary. This would likely require extensive ground testing, and potentially instrumented flight testing to gain insight into density effects. A P-Q (Pressure and Flow Rate) measurement device could be used to characterize the system with better accuracy than the previous methods. Pressure transducers measuring pressure drop across the system would provide a more primitive, better property to evaluate the state of system than dynamic pressure alone. With these data, one could more confidently define a control law to optimize the slowest air flow rate for a desired measurement frequency.

Chapter 10: References

[1] University of Groningen Mathematical Modeling of Complex Systems. (n.d.-c). https://pure.rug.nl/ws/portalfiles/portal/3207775/thesis.pdfyp

[2] Galewsky J, Steen-Larsen HC, Field RD, Worden J, Risi C, Schneider M. Stable isotopes in atmospheric water vapor and applications to the hydrologic cycle. Rev Geophys. 2016 Dec;54(4):809-865. doi: 10.1002/2015rg000512. Epub 2016 Aug 22. PMID: 32661517; PMCID: PMC7357203.

[3] Flight Operations Manual Flight Operations. (n.d.). https://www.colorado.edu/publicsafety/media/258

[4] J 3 1176 00067 9044 ,- :CHNICAL memorandum 81868 NASA-TM-81868 19810020599. (n.d.-b). https://ntrs.nasa.gov/api/citations/19810020599/downloads/19810020599.pdf

[5] *Intro to FreeCAD Part 4: How to Use Heat Set Inserts*. (2025). DigiKey. https://www.digikey.com/en/maker/tutorials/2025/intro-to-freecad-part-4-how-to-use-heat-set-inserts

[6] Wein, E. D., Kalnajs, L. E., & Toohey, D. W. (2024). An economical tunable diode laser spectrometer for fast-response measurements of water vapor in the atmospheric boundary layer. *Atmospheric Measurement Techniques*, *17*(24), 7097–7107. https://doi.org/10.5194/amt-17-7097-2024

[7] *Summit Camp climate: weather by month, temperature, rain - Climates to Travel*. (2021). Climatestotravel.com. https://www.climatestotravel.com/climate/greenland/summit-camp#google_vignette

[8] Arduino, *"Arduino Nano – Technical Specifications,"* Available: https://docs.arduino.cc/hardware/nano/#tech-specs. Accessed: 1 Dec. 2025.

[9] Vaisala, *"HMP60 Humidity and Temperature Probe Datasheet (B210851EN-K),"* Available: https://docs.vaisala.com/v/u/B210851EN-K/en-US. Accessed: 1 Dec. 2025.

[10] Sanyo Denki, *"San Ace E General Catalog,"* Available: https://publish.sanyodenki.com/library/books/San_Ace_E/book/#target/page_no=40. Accessed: 1 Dec. 2025.

[11] West System Gflex. (2025, December 1). *WEST SYSTEM G/flex Epoxy (2- 4oz)* [Review of *WEST SYSTEM G/flex Epoxy (2- 4oz)*]. Amazon.com. WEST SYSTEM G/flex Epoxy (2-4oz)

[12] *Printed Epoxy Bonding*. (2022). Jcrocket.com. https://jcrocket.com/printed-bonding.shtml

[13] Shin, H. H., Hong, S., & Dudhia, J. (2012). "*Impacts of the Lowest Model Level Height on the Performance of Planetary Boundary Layer Parameterizations,"* Monthly Weather Review, 140(2), 664-682. https://doi.org/10.1175/MWR-D-11-00027.1)

[14] "*The parametrization of the planetary boundary layer,"* Anton Beljaars, May 1992, European Centre for Medium-Range Weather Forecasts

[15] Equivalent potential temperature - glossary of meteorology. (n.d.-a). https://glossary.ametsoc.org/wiki/Equivalent_potential_temperature